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Research paper

HXAI-ML: A hybrid explainable artificial intelligence based machine learning model for cardiovascular heart disease detection

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ABSTRACT

Cardiovascular diseases (CVDs) are a leading cause of morbidity and mortality globally. Early diagnosis and accurate prediction are critical for effective prevention and treatment. However, traditional machine learning (ML) models for CVD prediction face challenges such as data imbalance, lack of interpretability, and limited generalization across datasets, which restrict their practical application in healthcare. This study introduces a hybrid explainable artificial intelligence-based ML (HXAI-ML) model to address these limitations. The proposed framework combines advanced data balancing techniques, including Random Oversampling (RO), Synthetic Minority Oversampling Technique (SM), RO+Tomek Link (TL), SM+TL, RO+Instant Hardness Threshold (IHT), and SM+IHT, with XAI tools like SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and Permutation Importance Analysis (PIA). This integration enhances prediction accuracy while ensuring model transparency, allowing healthcare professionals to interpret decision-making processes effectively. The HXAI-ML model was validated on two benchmark datasets, Cleveland and Framingham, achieving superior performance. Combining TL and RO with the Extra Trees Classifier (ETC) yielded the highest results, with 98.78% accuracy and 98.82% precision on the Cleveland dataset, and 99.24% accuracy and precision on the Framingham dataset. These findings demonstrate the model's ability to outperform traditional approaches in both accuracy and interpretability. This research provides a scalable, clinically interpretable solution for early CVD detection, capable of improving decision-making and enabling personalized treatment strategies. The hybrid model represents a significant step toward integrating advanced ML into practical healthcare applications, improving patient outcomes.

1. Introduction

CVDs, including CHD, stroke, heart failure, and peripheral artery disease, continue to be the leading causes of mortality worldwide [1]. According to the WHO, CVDs account for approximately 18 million deaths annually, representing nearly one-third of all global deaths [2,3]. The prevalence of CVDs has significantly increased due to various risk factors, including hypertension, diabetes, smoking, unhealthy diets, and a lack of physical activity [4]. As urbanization, aging populations, and lifestyle changes continue to affect populations globally, the burden of CVDs is expected to rise further, making it one of the most pressing public health issues [5]. The impact of CVDs extends beyond individual health, contributing significantly to healthcare costs, economic loss, and reduced quality of life [6]. Early identification of individuals at risk is crucial for effective intervention, as timely treatment and lifestyle modifications can substantially lower the incidence of heart attacks, strokes, and other related complications [7,8]. Prevention strategies such as controlling blood pressure, managing cholesterol levels, and promoting healthier lifestyles are essential for reducing the prevalence of CVDs [9]. However, despite these preventive measures, many individuals remain undiagnosed or misdiagnosed until they experience an acute cardiovascular event, often with severe consequences [10]. Therefore, there is a growing need for more effective and precise diagnostic tools that can predict CVD risk accurately, enabling better prevention and timely treatment interventions at an early stage.

Traditional risk prediction models, such as the FHS, have played a pivotal role in shaping our understanding of cardiovascular risk factors [11,12]. These models use predefined clinical parameters, such as age,

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blood pressure, cholesterol levels, and smoking history, to estimate the likelihood of a person developing CVD over a given time frame [13]. The Framingham model and similar traditional methods have provided valuable insights into the identification of high-risk individuals and have been widely used in clinical practice for decades [14]. However, despite their significant contributions to CVD risk prediction, these models have notable limitations. Most traditional models, including the FHS, rely on linear relationships between risk factors, which can oversimplify the complex, non-linear interactions that characterize real-world clinical situations [15]. The interactions among various risk factors, such as the combination of hypertension with diabetes or smoking with cholesterol levels, can influence the development of CVD in ways that linear models are unable to capture [16]. As a result, these models often struggle to provide accurate predictions for individual patients, especially in diverse populations with multifactorial risk factors. Moreover, traditional models do not account for emerging risk factors or dynamic changes in a patient's health over time, reducing their applicability in modern clinical settings [17].

Furthermore, imbalanced datasets—where the number of positive cases is significantly lower than negative cases—further complicate the effectiveness of these models. Lastly, the lack of interpretability in predictions generated by traditional models makes it challenging for healthcare professionals to trust and apply the results in clinical decision-making, where transparency and justifiability are essential [18–20]. These challenges highlight the need for more advanced and flexible approaches, such as ML and XAI, to overcome the limitations of traditional risk prediction models and provide more personalized and accurate CVD risk assessments. ML has gained significant attention in CVD prediction due to its ability to handle large, complex datasets and model non-linear relationships among risk factors [21,22]. ML techniques offer the potential to improve prediction accuracy over traditional models [23,24]. These can also identify subtle patterns and correlations within patient data that may be overlooked by conventional approaches [25,26]. However, issues such as imbalanced datasets, generalizability, and the need for model interpretability remain critical challenges in deploying ML models for CVD prediction in clinical practice. To overcome these hurdles, advanced data preprocessing, robust algorithms, and explainable AI techniques are necessary to ensure the reliability and transparency of ML-based models in healthcare.

1.1. Technical challenges

While the healthcare problem is critical, addressing the technical challenges associated with machine learning (ML) models for CVD prediction is equally important. These challenges include:

- Data Imbalance: CVD datasets often have a significantly lower number of positive cases (CVD presence) compared to negative cases, resulting in biased predictions and limited generalization across diverse populations.
- Lack of Interpretability: Many ML models function as black boxes, making it difficult for clinicians to trust or adopt them in practice, where explainability is crucial for transparent decision-making.
- Limited Generalization: ML models frequently struggle to maintain consistent performance across datasets with varying data distributions, reducing their scalability and real-world applicability.

To overcome these challenges, the proposed research introduces a Hybrid Explainable Artificial Intelligence-Based Machine Learning (HXAI-ML) framework. This framework incorporates advanced hybrid data balancing techniques and explainable AI tools to enhance prediction accuracy, improve interpretability, and ensure scalability. By addressing these limitations, the HXAI-ML framework offers a robust and clinically viable solution for accurate and interpretable CVD prediction in real-world healthcare applications.

1.2. Problem statements

Despite significant advancements in ML-based models for CVD prediction, several challenges remain:

- Data Imbalance: CVD datasets are often imbalanced, with fewer samples for positive cases, leading to biased predictions and poor model generalization.
- Interpretability: The lack of transparency in ML predictions limits their adoption in clinical settings where decisions must be interpretable and justifiable.
- Limited Hybrid Approaches: Existing works often use either traditional balancing techniques or ML algorithms in isolation, missing opportunities to combine strengths through hybrid approaches.

In response to these challenges, ML and XAI techniques are revolutionizing the field of cardiovascular health prediction. By leveraging advanced data-driven approaches, ML models can enhance prediction accuracy, scalability, and the interpretability of clinical outcomes. However, existing works often struggle to address imbalanced datasets comprehensively or fail to incorporate robust XAI techniques, which are essential for clinical applicability.

1.3. Objectives

This research aims to address these challenges with the following objectives:

- Develop a novel hybrid ML model for accurate CVD prediction that addresses data imbalance and improves interpretability.
- Evaluate advanced hybrid data balancing techniques combining oversampling and undersampling methods.
- Incorporate XAI techniques to ensure the proposed model provides clinically interpretable predictions.

1.4. Contributions

The key contributions of this study are as follows:

- Novel Hybrid Model: Developed a hybrid XAI-based ML model (HXAI-ML) for CVD detection using advanced data balancing and interpretable techniques.
- Advanced Data Balancing: Introduced a comprehensive hybrid data balancing approach combining RO, SM, TL, and IHT. Combinations such as RO+TL, RO+IHT, SM+TL, and SM+IHT were evaluated to identify the optimal strategy for addressing data imbalance.
- Explainable AI Integration: Incorporated XAI techniques, including PIA, SHAP, and LIME, to enhance model transparency and provide interpretable predictions for clinical applications.

1.5. Research questions

- What is the most effective hybrid data balancing technique for improving model performance in imbalanced CVD datasets?
- How can XAI techniques like SHAP, LIME, and PIA enhance the interpretability of CVD prediction models in clinical settings?
- Does the proposed hybrid model outperform existing ML-based methods in terms of accuracy, sensitivity, and precision?

The remainder of this paper is organized as follows: Section 2 provides a review of related work on CVD prediction. Section 3 details the proposed methodology, including a comprehensive description of the approach. Section 4 presents the performance analysis of the proposed method. Section 5 compares the proposed approach with existing works, addresses the validation of the research questions, and discusses key

findings. Finally, Section 6 concludes the paper by summarizing the research contributions, highlighting limitations, and suggesting directions for future work.

2. Related works

CVD prediction has been extensively studied using various datasets, including the Cleveland and Framingham datasets. Researchers have explored a wide range of ML and data-driven approaches to enhance predictive accuracy and address the limitations of traditional models. The following discusses the related works based on these datasets and the advancements made in CVD prediction.

2.1. Related works on Cleveland dataset

Explainable AI (XAI) methods Mienye and Jere [27] proposed an optimized ensemble learning approach for heart disease prediction, integrating SHAP for interpretability. XGBoost, optimized via Bayesian hyperparameter tuning, achieved 98.4% accuracy on Cleveland and 95.9% on Framingham datasets, demonstrating the effectiveness of combining ensemble learning with explainability for improved clinical predictions.

Feature selection techniques Ranade et al. [28] investigated feature selection techniques for cardiovascular disease prediction using a hybrid ML model on the UCI Cleveland dataset. The LR model with SFS achieved the highest accuracy of 84.4%, highlighting the role of feature selection in enhancing ML model performance. Shokouhifar et al. [29] introduced the EHMFFL algorithm for heart disease diagnosis, combining ensemble learning with heuristic–metaheuristic optimization. Using Pearson correlation and the grey wolf optimizer for feature selection, and aggregating SVM, KNN, and XGBoost outputs, EHMFFL achieved 91.8% and 88.9% accuracy on Cleveland and Statlog datasets, respectively, outperforming existing methods and aiding clinical decision-making. Sivakami and Prabhu [30] compared classification algorithms for heart disease prediction, achieving 96% accuracy by combining linear regression with decision trees on a diabetic-related heart disease dataset. Their work underscored the effectiveness of ML in handling medical data and the importance of feature selection in reducing complexity. Al-Tashi et al. [31] proposed a GWO-based feature selection method for coronary artery disease prediction. Using GWO-SVM, they achieved 89.83% accuracy, with 93% sensitivity and 91% specificity, demonstrating GWO's superiority over traditional optimization methods for feature selection in ML. Perumal et al. [32] focused on early coronary heart disease prediction using PCA for feature reduction and classifiers such as KNN, SVM, and LR. On the Cleveland dataset, LR achieved the highest accuracy of 87%, demonstrating that reducing data dimensionality can maintain high prediction performance. Elsevimy et al. [33] developed QPSO-SVM, a heart disease detection model that integrates QPSO with SVM for feature selection. Optimized with a self-adaptive threshold to balance exploration and exploitation, QPSO-SVM achieved 96.31% accuracy on the Cleveland dataset, outperforming other models in sensitivity, specificity, precision, and F1 score.

CVD detection using ML models Nazari et al. [34] introduced a novel ensemble model for predicting cardiovascular diseases (CVD) and identifying key factors using the Cleveland and Noor datasets. Combining classifiers with weights optimized via a genetic algorithm, the model achieved precision rates of 88.05% and 90.12%, respectively. This approach advanced early detection and understanding of CVD. Mienye et al. [35] proposed an improved ensemble method for heart disease prediction, leveraging a modified CART approach with mean-based data splitting and AB-WAE optimization. Their model achieved 93% and 91% accuracies on the Cleveland and Framingham datasets, demonstrating EL's efficacy in medical risk prediction. Chandrasekhar et al. [36] evaluated six ML algorithms for heart disease prediction and achieved 93.44% accuracy on the Cleveland dataset using a soft-voting ensemble classifier. The study highlighted the superiority of ensemble methods for medical predictions. Anderies et al. [37] analyzed heart disease prediction on the UCI Cleveland dataset, finding SVM to have the highest accuracy of 85%. Their work emphasized EL's role in improving precision in healthcare diagnostics. Srinivas et al. [38] proposed the hyOPTXg framework, using OPTUNA for hyperparameter tuning of XGBoost. Achieving 94.7% accuracy on the Cleveland dataset, this framework demonstrated scalability and robustness in heart disease prediction. Chhillar et al. [39] compared eight ML algorithms on the Statlog and Cleveland datasets, identifying SVM as the most accurate (84.15%) on Cleveland, while LR excelled in ROC-AUC. Their findings emphasized optimizing ML models for healthcare applications. Akella et al. [40] applied six ML models, including NN, RF, and SVM, to coronary artery disease prediction. NN outperformed others, achieving 93% accuracy, demonstrating its strength in handling complex medical datasets.

CVD detection using DL models Kayalvizhi et al. [41] proposed an Optimized CNN-BiLSTM model with attention mechanisms for CVD detection, achieving 98.1% accuracy on Framingham and 95.3% on Cleveland datasets. By integrating CNN for spatial features, BiLSTM for temporal dependencies, and NRO for parameter tuning, the model showcased robust prediction capabilities.

2.2. Related works on framingham dataset

Data balancing techniques Kahouadji et al. [42] performed a comparative study on several ML classifiers, including BDT, SVM, and LR, across three variations of the Framingham Heart Study dataset. BDT emerged as the best performer with an accuracy of 85% on the unprocessed dataset, effectively managing class imbalances using ROSE techniques. Louridi et al. [43] developed an ML-based system for cardiovascular disease diagnosis utilizing stacking algorithms. Achieving a 90.24% accuracy on the Framingham dataset, the system leveraged MICE and RF for data imputation and SM for class imbalance handling. Their study highlighted the critical role of preprocessing in boosting prediction accuracy. Rahim et al. [44] proposed the MaLCaDD framework, integrating LR and KNN classifiers for CVD prediction. The framework achieved 99.1% accuracy on the Framingham dataset and 95.5% on the Cleveland dataset. By employing SM for data balancing and optimizing feature selection, MaLCaDD demonstrated exceptional accuracy and efficiency, proving valuable for early CVD diagnosis.

Explainable AI (XAI) methods Sunge et al. [45] proposed a heart disease prediction model using SHAP for feature selection and interpretability. Testing on the Framingham dataset, they evaluated classifiers like LR, KNN, DT, RF, and GB. Both GB and LR achieved 85% accuracy, with key predictors identified as age, systolic blood pressure, and cigarettes per day.

Feature selection techniques Gupta et al. [46] introduced a GA-based method for feature selection and hyperparameter optimization on the Framingham Heart Disease dataset. Optimizing SVM with an RBF kernel and MLP, they achieved accuracies of 83% and 82%, respectively, highlighting GA's role in boosting prediction performance.

CVD detection using ML models Omotehinwa et al. [47] employed Bayesian optimization to enhance the LightGBM algorithm for CHD detection. Their optimized model achieved an accuracy of 98.82% and an AUC of 99.63%, demonstrating the effectiveness of LightGBM with advanced hyperparameter tuning. Suhatril et al. [48] assessed several ML models, including RF, NB, and SVM, on the Framingham dataset for cardiovascular disease prediction. RF achieved the highest accuracy of 85%, while NB recorded the best AUC of 0.72, underlining the utility of ML in healthcare predictions. Zhang et al. [49] analyzed genome-wide DNA methylation data from the Framingham dataset to identify biomarkers for CHD. Using LightGBM, XGBoost, and RF, they achieved

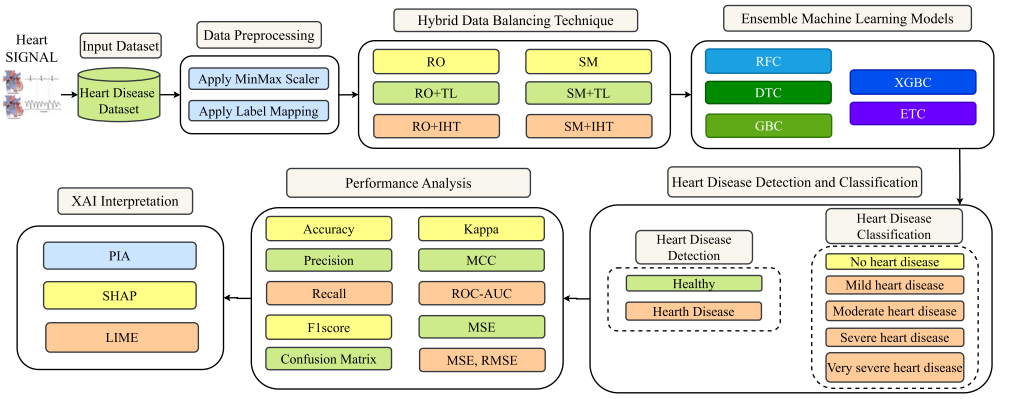


Fig. 1. Proposed HXAI-ML Approach for Heart Disease Detection.

an AUC of 83.4% with LightGBM, emphasizing the integration of genetic data for improved CHD risk prediction. Guarneros et al. [50] developed ML models for cardiovascular risk prediction, achieving accuracies of 77.42% with LGBM on the Framingham dataset and 77.22% with LR on the Cleveland dataset. Their work highlights the importance of key feature selection for improving diagnostic accuracy in resource-constrained environments. Nalluri et al. [51] compared XGBoost and LR for CHD prediction on the Framingham dataset. LR achieved the highest accuracy of 85.86%, while XGBoost reached 84.46%. Their findings demonstrated that LR excels with smaller datasets, whereas XGBoost offers faster processing for larger ones. Beunza et al. [52] evaluated various ML models, such as DT, RF, SVM, NN, and LR, for coronary heart disease prediction using the Framingham dataset. SVM recorded the highest AUC of 75%, and LR achieved the best accuracy of 69%, showcasing the potential of ML in predicting coronary events.

CVD detection using DL models Gupta et al. [53] compared ML and DL models for heart disease prediction. Their results showed that the RF model achieved the highest accuracy of 97.13% on the Framingham dataset, surpassing other models. This study highlighted the significant improvements in heart disease prediction accuracy through hyperparameter tuning and DL models. Demir et al. [54] employed KNN and ANN to predict coronary heart disease using the Framingham dataset. They used the Hotdeck imputation method to handle missing data and found that ANN outperformed KNN, achieving an accuracy of 85.85%, while KNN reached 85.50%. Their study emphasizes the advantage of ANN in achieving higher prediction accuracy for heart disease diagnosis. The summary of the CHD related works are represented in Table 1.

While numerous studies have advanced cardiovascular disease prediction using ML techniques and XAI tools, several critical limitations persist. Traditional approaches often fail to address challenges such as data imbalance, which can skew predictions and lack the transparency required for clinical applications. Recent advancements, such as ensemble learning and hybrid algorithms, have shown improvements in accuracy. However, these methods are frequently limited by insufficient integration of advanced data balancing techniques and comprehensive interpretability frameworks. Our proposed HXAI-ML framework bridges these gaps by incorporating hybrid data balancing techniques with state-of-the-art XAI tools, ensuring not only improved predictive performance but also enhanced interpretability for clinical decision-making. This comprehensive approach surpasses existing methodologies by providing a scalable and reliable solution tailored to real-world healthcare scenarios. By addressing these critical gaps, our work transitions seamlessly into the methodology section, where we detail the implementation and evaluation of the HXAI-ML framework.

3. Proposed methodology

The proposed HXAI-ML approach aims to provide an effective and interpretable solution for heart disease detection by combining ML techniques with XAI (Fig. 1). The process begins with the input of heart disease dataset, which includes various medical features such as ECG signals and patient characteristics. The dataset undergoes essential preprocessing, which involves scaling the data using the MinMax Scaler to ensure consistency and applying label mapping for numerical to categorical variables. To address the issue of class imbalance, the methodology incorporates a hybrid data balancing technique, which combines several over-sampling and under-sampling methods. These techniques include RO, TL, RO+TL, RO+IHT, RO+IHT and SM+IHT to ensure that the model is trained on a balanced dataset, reducing bias toward the majority class. This balancing is critical for improving the model's ability to classify both healthy and diseased cases accurately.

The heart disease classification is then performed using ensemble ML models such as RFC, DTC, GBC, and XGBC. These models are chosen for their high accuracy and robustness in handling complex patterns within the data. HXAI-ML employs Explainable AI techniques such as SHAP, LIME, and PIA to interpret the decisions made by the models, providing insights into the most influential features in heart disease detection. The performance analysis evaluates the models using various metrics such as accuracy, precision, recall, F1-score, ROC-AUC, Kappa, MCC, and RMSE. These metrics ensure a comprehensive assessment of the model's predictive performance. The final classification includes different heart disease stages, ranging from "No Heart Disease" to "Very Severe Heart Disease," enabling the system to not only detect heart disease but also classify its severity, providing valuable information for clinical decision-making. This integrated approach of HXAI-ML offers not only high prediction accuracy but also interpretability, which is crucial for healthcare applications where model transparency is essential.

3.1. Flow of the HXAI-ML framework

The flow of the proposed HXAI-ML Framework for heart disease detection is outlined in Fig. 2. The framework is designed to ensure accurate, interpretable, and clinically meaningful predictions by combining advanced data balancing techniques, robust ML models, and XAI tools. The step-by-step methodology is detailed below:

- **Dataset Acquisition:** The process begins with the collection of a heart disease dataset, which includes essential medical features such as ECG signals, patient demographics, and clinical test results. These datasets form the foundation of the HXAI-ML framework, enabling the system to model complex relationships between features and disease outcomes.

Table 1
Summary of CHD Prediction Models and Performance Metrics.

Study	Dataset	Approach	Techniques Used	Accuracy	Precision	Interpretability	Key Findings
Ranade et al. [28]	Cleveland	Feature Selection	Hybrid Model, ML Algorithms, SFS	84.4%	62.5%	-	Emphasized feature selection's role in enhancing ML model performance.
Shokouhifar et al. [29]	Cleveland, Statlog	Ensemble Learning	EL, SVM, KNN, XGBoost, Grey Wolf Optimizer	91.8% (Cleveland)	91.4%	High	Optimized feature selection for base learners, significantly improving diagnostic accuracy.
Sivakami and Prabhu [30]	Cleveland	Classification	NB, DT, KMC, LR	98.35%	98.4%	Medium	Focused on classification algorithms for heart disease prediction, achieved high accuracy on a diabetes-related heart disease dataset.
Al-Tashi et al. [31]	Cleveland	Feature Selection	GWO, SVM	89.83%	-	High	Highlighted the effectiveness of GWO for feature selection in coronary artery disease prediction.
Mienye and Jere [27]	Cleveland, Framingham	Explainable AI	SHAP, RF, AdaBoost, XGBoost	98.4% (Cleveland), 95.9% (Framingham)	-	High	Demonstrated the power of XGBoost combined with SHAP for interpretability and accuracy.
Kayalvizhi et al. [41]	Cleveland, Framingham	DL	CNN-BiLSTM, Attention Mechanism	95.3% (Cleveland)	-	Medium	Emphasized CNN and BiLSTM integration for enhanced prediction.
Nazari et al. [34]	Cleveland, Noor	Ensemble Learning	Genetic Algorithm, EL	88.43%	88.05%	High	Introduced a novel ensemble method to identify key contributing factors for CVD prediction.
Elsevimy et al. [33]	Cleveland	Feature Optimization	QPSO, SVM	96.31%	94.23%	Medium	Developed QPSO-SVM, which outperformed traditional models in heart disease detection.
Chandrasekhar et al. [36]	Cleveland	Ensemble Learning	RF, KNN, LR, NB, GB, AdaBoost	93.44%	88%	Low	Demonstrated the effectiveness of soft-voting ensemble classifiers for improved CVD prediction.
Anderies et al. [37]	Cleveland	Classification	NB, SVM	85%	0.97%	Low	SVM outperformed other classifiers, emphasizing precision in heart disease diagnostics.
Srinivas et al. [38]	Cleveland	Hyperparameter Optimization	OPTUNA, XGBoost	94.7%	92.65%	Medium	Highlighted the importance of hyperparameter optimization in tree-based models.
Chhillar et al. [39]	Cleveland, Statlog	Classification	SVM, NB, DT, RF, LR	84.15% (Cleveland)	84.3%	Low	Showcased the need for optimizing classifiers for enhanced healthcare predictions.
Akella et al. [40]	Cleveland	Classification	NN, RF, SVM, Others	93%	-	Medium	NN demonstrated high performance in handling complex medical datasets.
Perumal et al. [32]	Cleveland	Dimensionality Reduction, Classification	PCA, KNN, SVM, LR	87%	-	Medium	PCA helped reduce data complexity while maintaining high prediction accuracy.
Mienye et al. [35]	Cleveland, Framingham	Explainable AI	CART, AB-WAE	93% (Cleveland), 91% (Framingham)	96%	Medium	Highlighted the success of enhanced CART-based ensemble learning for heart disease prediction.
Kahouadji et al. [42]	Framingham	Classification	BDT, SVM, LR, ROSE	85%	84.95%	Medium	Emphasized the importance of class balancing and model selection for improved performance.
Louridi et al. [43]	Framingham	Data Imputation, Classification	Stacking, MICE, RF	90.24%	92%	Medium	Integrated multiple preprocessing techniques for handling missing data and class imbalance.
Rahim et al. [44]	Cleveland, Framingham	Ensemble Learning	LR, KNN	99.1% (Framingham)	-	Medium	Achieved very high accuracy by addressing data imbalance and optimizing feature selection.
Sunge et al. [45]	Framingham	Explainable AI	SHAP, LR, KNN, DT, RF, GB	85%	85%	High	Integrated SHAP for feature interpretability, with good accuracy using LR and GB models.
Omotehinwa et al. [47]	Framingham	Hyperparameter Optimization	LightGBM, Bayesian Optimization	98.82%	98.35%	High	Optimized LightGBM model showed impressive accuracy and AUC performance.
Suhatril et al. [48]	Framingham	Classification	RF, NB, SVM	84%	84%	Medium	RF outperformed others in terms of accuracy, while NB showed the best AUC.

(*continued on next page*)

Table 1 (*continued*)

Study	Dataset	Approach	Techniques Used	Accuracy	Precision	Interpretability	Key Findings
Demir et al. [54]	Framingham	Classification	KNN, ANN	85.85% (ANN)	99.58%	Low	ANN outperformed KNN, highlighting the strength of NN for heart disease prediction.
Gupta et al. [53]	Framingham	DL, ML	RF, XGBoost, ANN	97.13%	96%	Low	RF achieved highest accuracy, emphasizing the role of hyperparameter tuning in DL.
Zhang et al. [49]	Framingham	Genetic Data, ML	LightGBM, XGBoost, RF	83.4% AUC (LightGBM)	61.6%	High	Integrated genomic data for more sensitive and specific heart disease prediction.
Nalluri et al. [51]	Framingham	Classification	XGBoost, LR	85.86% (LR)	-	Medium	LR achieved better accuracy with smaller datasets, while XGBoost was more efficient with larger datasets.
Gupta et al. [46]	Framingham	Genetic Data, Feature Selection	GA, SVM	83%	87%	Medium	Used genetic algorithms for feature selection, optimizing model learning.

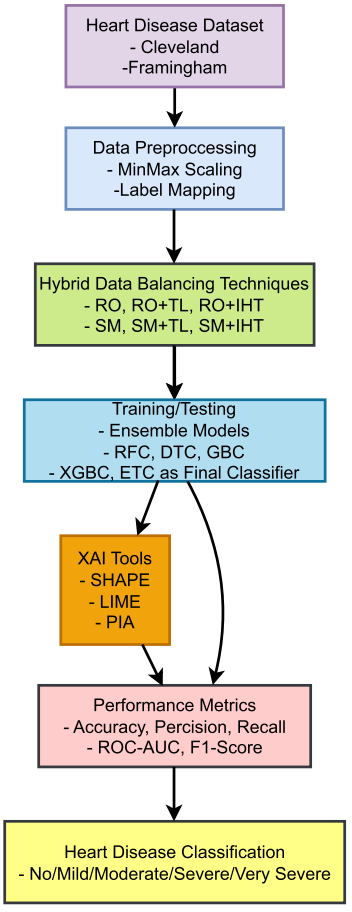


Fig. 2. The Flowchart of the proposed HXAI-ML Framework.

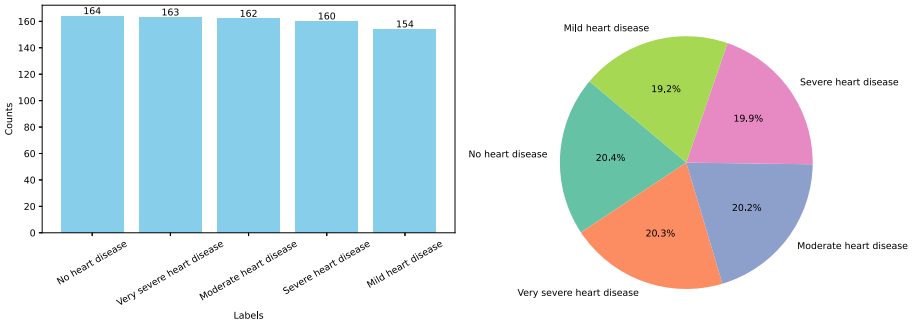
- **Preprocessing:** The dataset undergoes essential preprocessing to prepare it for machine learning analysis. Scaling is performed using the MinMax Scaler to normalize the feature values within a uniform range, ensuring consistency and improving model performance. Additionally, label mapping is applied to convert categorical variables into numerical representations, facilitating seamless integration with machine learning algorithms.
- **Data Balancing:** To address the issue of class imbalance, the framework employs a hybrid data balancing technique that combines oversampling and undersampling methods. These include RO, TL, SM, and IHT. Various combinations, such as RO+TL, RO+IHT and SM+RO, SM+IHT, are tested to identify the optimal strategy. This step ensures the dataset is balanced, enabling the models to classify both majority and minority classes accurately.
- **Model Training:** Ensemble machine learning models are used to classify heart disease stages. These include RFC, DTC, GBC, and XGBC. The models are trained to predict whether a patient is healthy or suffers from one of several stages of heart disease, ranging from "No Heart Disease" to "Very Severe Heart Disease."
- **XAI Integration:** To ensure interpretability and transparency, the HXAI-ML framework integrates Explainable AI techniques:
  - **SHAP:** Quantifies the contribution of each feature to the model's predictions.
  - **LIME:** Provides local explanations for individual predictions, helping to understand specific decisions.
  - **PIA:** Highlights the global impact of features across the dataset, making the predictions clinically relevant.
- **Performance Evaluation:** The performance of the HXAI-ML framework is evaluated using a comprehensive set of metrics, including accuracy, precision, recall, F1-score, ROC-AUC, Kappa, MCC, and RMSE. These metrics provide a holistic assessment of the model's predictive capabilities, ensuring reliability and robustness for clinical applications.
- **Final Outcome:** The HXAI-ML framework outputs both the classification of heart disease stages and interpretable insights into the most influential features driving the predictions. This dual output supports clinical decision-making, enabling healthcare professionals to make informed and evidence-based decisions.

The proposed HXAI-ML framework presents a structured and integrated approach to heart disease detection, leveraging advanced data balancing techniques, ensemble machine learning models, and XAI tools. This comprehensive design ensures both high predictive accuracy and interpretability, making it a practical and reliable solution for healthcare applications.

3.2. *Data preprocessing*

In our experiment, both the Cleveland Heart Disease Dataset and the Framingham Heart Disease Dataset were observed to have no missing values, and all features were represented as numerical values. To prepare the datasets for ML algorithms and ensure consistent performance evaluation, we applied MinMax scaling for feature normalization and label mapping for clarity in evaluating classification metrics.

Label mapping While the target columns in both datasets were initially numerical, we performed label mapping during the performance evaluation process to enhance interpretability in metrics such as confusion matrices and classification reports. Specifically, the numerical values were mapped to descriptive categorical labels:



(a) The Count of Each Label (b) The Percentage of Each Label

Fig. 3. Data Distribution after Data Balancing using RO+TL.

- *Cleveland Dataset:*
  - 0 → "No heart disease"
  - 1 → "Mild heart disease"
  - 2 → "Moderate heart disease"
  - 3 → "Severe heart disease"
  - 4 → "Very severe heart disease"
- *Framingham Dataset:*
  - 0 → "Healthy"
  - 1 → "Heart Disease"

This mapping ensures clarity in analyzing and interpreting performance results while maintaining compatibility with the ML framework.

By applying these preprocessing steps, the datasets were standardized and made suitable for input into the proposed hybrid model while ensuring effective performance evaluation.

3.3. Data balancing techniques

To address class imbalance in our datasets, we employed a combination of data-balancing techniques. These hybrid techniques enhance the distribution of the minority and majority classes, ensuring that the models are trained on balanced data. In particular, we experimented with the following methods:

Random oversampling (RO) RO involves duplicating instances from the minority class until the classes are balanced. While effective in mitigating imbalance, it can lead to overfitting due to repeated data points.

Synthetic minority oversampling technique (SM) SM generates synthetic samples for the minority class by interpolating between existing instances. It creates new samples based on the feature space, reducing overfitting associated with random oversampling. The synthetic sample creation is defined as:

$$x_{\text{new}} = x_{\text{min}} + \lambda \cdot (x_{\text{neighbor}} - x_{\text{min}}) \tag{1}$$

where:

- x_{new} is the synthetic instance,
- x_{min} is a minority class instance,
- x_{neighbor} is a randomly selected neighbor of x_{min},
- λ is a random number between 0 and 1.

Random oversampling with Tomek links (RO+TL) Our proposed approach combines RO with TL where TL identify PIAr of instances from different classes that are nearest neighbors and whose removal reduces class overlap. The hybrid technique first applies random oversampling to balance the data, followed by TL cleaning. Let x_i and x_j be a PIAr of instances belonging to different classes; they form a TL if:

$$d(x_i, x_j) < d(x_i, x_k) \quad \text{and} \quad d(x_j, x_i) < d(x_j, x_l) \tag{2}$$

where:

- $d(x_i, x_j)$ is the distance between x_i and x_j,
- x_k and x_l are instances from other classes.

This method reduces overlapping instances, enhancing decision boundaries.

SMOTE with Tomek links (SM+TL) This hybrid method combines SM for generating synthetic minority samples and TL for cleaning class overlaps. It balances the data while ensuring better separation between classes.

Random oversampling with instance hardness threshold (RO+IHT) IHT measures the difficulty of classifying instances based on the probability of correct classification by the model. The RO+IHT approach applies random oversampling and removes hard-to-classify instances using a threshold to refine the data distribution.

SMOTE with instance hardness threshold (SM+IHT) This hybrid method generates synthetic samples for the minority class using SM and eliminates instances with high classification difficulty based on IHT. It balances the data while reducing noisy and ambiguous samples.

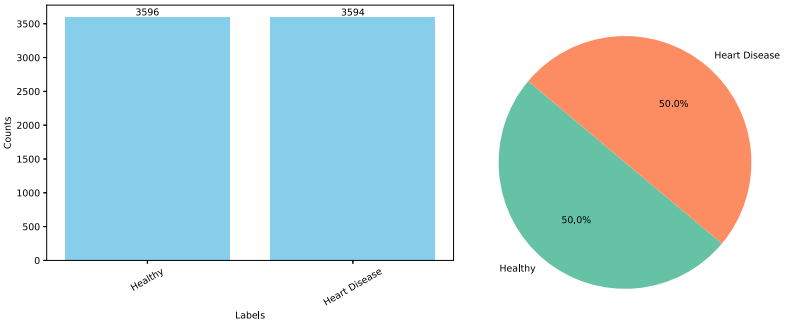
Proposed approach: RO+TL Among the above methods, our proposed approach, RO+TL, demonstrated superior performance in mitigating class imbalance and improving model generalization. By combining RO to balance the data and TL to clean overlaps, this method effectively prepares the datasets for ML algorithms.

Fig. 3 illustrates the balanced data distribution for the Cleveland Heart Disease Dataset after applying the RO+TL technique, showcasing evenly distributed classes in both count and percentage. Similarly, Fig. 4 depicts the improved class balance for the Framingham Heart Disease Dataset, achieved through the same method, ensuring a fair representation of all classes for model training.

3.4. Ensemble machine learning (EML) models

EML models combine the predictions of multiple individual models to achieve higher accuracy and robustness, particularly in complex tasks like CVD prediction. This subsection discusses the ensemble algorithms RFC, DTC, GBC, XGBC, and ETC, along with their benefits for CVD detection.

Random forest classifier (RFC): RFC is a bagging-based ensemble method that constructs multiple decision trees during training and aggregates their predictions, either by majority voting for classification or averaging for regression. The RFC algorithm introduces randomness by selecting a random subset of features for splitting at each node, which



(a) The Count of Each Label (b) The Percentage of Each Label

Fig. 4. Data Distribution after Data Balancing using RO+TL on Framingham Dataset.

helps reduce overfitting and improves generalization. Mathematically, the final prediction $\hat{y}$ of RFC is given by:

$$\hat{y} = \frac{1}{T}\sum_{t=1}^{T} h_t(x),$$

where $h_t(x)$ represents the prediction of the t-th tree and T is the total number of trees. RFC is particularly beneficial for CVD detection due to its ability to handle high-dimensional data, resilience to noise, and superior performance on imbalanced datasets.

Decision tree classifier (DTC): DTC is a tree-based model that recursively splits the dataset into subsets based on feature thresholds to maximize information gain or minimize impurity. The decision rules are derived from the features, making the model inherently interpretable. The impurity reduction at a node is often calculated using metrics like Gini impurity or entropy. For example, the Gini impurity at a node is:

$$G = 1 - \sum_{i=1}^{C} p_i^2,$$

where p_i is the proportion of class i instances, and C is the total number of classes. While simple, DTCs are effective for CVD detection because they capture non-linear patterns and provide transparent decision rules for clinical interpretation.

Gradient boosting classifier (GBC): GBC is a boosting technique that builds sequential models, where each tree corrects the errors of its predecessor. This iterative process minimizes a differentiable loss function, such as the log-loss for classification tasks. The update rule for the model prediction $\hat{F}(x)$ at iteration m is:

$$\hat{F}_m(x) = \hat{F}_{m-1}(x) + \eta \cdot h_m(x),$$

where η is the learning rate and $h_m(x)$ is the m-th base learner. GBC is highly effective for CVD detection as it captures complex interactions between features, offering improved predictive performance compared to standalone models.

Extreme gradient boosting classifier (XGBC): XGBC is an optimized implementation of gradient boosting that incorporates regularization, parallel processing, and efficient handling of missing data. The regularized objective function is:

$$\mathcal{L}(\theta) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \lambda \sum_{j=1}^{k} \omega_j^2,$$

where l is the loss function, $\hat{y}_i$ is the predicted value, ω_j are the model parameters, and λ is the regularization term. XGBC's ability to handle large datasets and model feature interactions with high precision makes it a powerful choice for detecting CVD.

Extra trees classifier (ETC): ETC is a variation of the random forest algorithm that introduces additional randomness by selecting splits randomly rather than optimizing them based on impurity measures. This randomness reduces overfitting and speeds up training. The prediction of ETC follows a similar averaging or majority-voting approach as RFC. ETC is particularly advantageous for CVD detection because of its computational efficiency and robustness to noisy data. By incorporating ETC into the proposed model, we expect to achieve superior predictive performance, especially for rare and severe CVD cases, while maintaining clinical interpretability. This aligns with the goal of developing a robust and transparent model that can aid in early detection and clinical decision-making.

Benefits of EML models in CVD detection: The EML models collectively offer improved accuracy, robustness, and scalability for CVD prediction. By combining multiple weak learners, these methods reduce the risk of overfitting and capture intricate patterns in medical datasets. The interpretability of tree-based models (DTC, RFC, ETC) and the high predictive performance of boosting methods (GBC, XGBC) enable a balance between accuracy and clinical relevance.

3.5. Explainable artificial intelligence (XAI) techniques

The XAI models integrated into the HXAI-ML model play a crucial role in enhancing the transparency and interpretability of predictions [55,56]. By incorporating techniques such as PIA, SHAP, and LIME, the HXAI-ML model ensures that both global and local feature contributions are clearly understood [57]. These methods enable clinicians to trust the model's decisions by providing detailed insights into the most influential factors driving the prediction of CVD, thereby improving its applicability in clinical settings. We selected SHAP, LIME, and PIA as the three chosen XAI methods for the following reasons:

SHAP (Shapley additive exPlanations): SHAP is widely recognized for its ability to provide consistent, mathematically sound feature importance values. It is based on cooperative game theory, where features are treated as "players" contributing to the model's prediction [58]. SHAP ensures that each feature's contribution is fairly distributed, which provides an accurate and interpretable explanation of model predictions. Its ability to handle both global and local interpretability makes it highly suitable for understanding complex ML models, particularly in sensitive applications like medical diagnosis. We prefer SHAP over other methods due to its consistency and the well-established theoretical foundation that makes it robust in providing fair and accurate insights into feature contributions.

LIME (local interpretable model-agnostic explanations): LIME is valuable for generating localized explanations, where it approximates the behavior of a complex model using simpler, interpretable models in the vicinity of specific predictions [59]. This makes LIME particularly useful for understanding individual predictions and providing insights into

Table 2
System Specifications and Experimental Setup.

Aspect	Details
Datasets Characteristics	Cleveland Heart Disease (303 samples, 14 attributes, 5 classes) and Framingham Heart Disease (4,240 samples, 16 attributes, 2 classes)
Preprocessing	MinMax Scaling, Label Mapping
Train-Test Splits	80% Training - 20% Testing
Data Balancing	RO, SM, RO+TL, SM+TL, RO+IHT, SM+IHT
ML Algorithms	RFC, DTC, GBC, XGBC, ETC
XAI Techniques	SHAP, LIME, PIA
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, MCC, Kappa, ROC-AUC, MAE, MSE, RMSE, Confusion Matrix and Classification Reports
Hardware Specifications	Intel Core i7 (10th Generation) processor, 16 GB RAM, NVIDIA GeForce GTX 1650 GPU (4 GB VRAM), Windows 10 Pro
Software Libraries	Scikit-learn: Preprocessing, training, and evaluating ML models NumPy/Pandas: Data manipulation and analysis Matplotlib/Seaborn: Data visualization (distributions and results) Imbalanced-learn: Data-balancing techniques

how a model arrived at a particular decision. It is particularly advantageous when individual case interpretations are critical, as in clinical settings where doctors need to understand the reasons behind a specific diagnosis. LIME's model-agnostic nature and the ability to focus on specific instances rather than the entire dataset make it a preferred choice for explaining individual predictions in real-world applications.

Permutation importance analysis (PIA): PIA is a simple yet effective method to assess feature importance by evaluating the impact of permuting each feature on the model's performance [60]. It helps to highlight the most influential features in a straightforward manner, which is crucial for establishing the relevance of features in a medical context. PIA is computationally less expensive compared to other methods like SHAP and LIME, and its ability to work directly with model performance metrics (e.g., accuracy, F1 score) ensures it provides clear and interpretable results for practitioners. We chose PIA because of its simplicity, ease of use, and ability to provide a global view of feature importance, complementing the localized insights from LIME and the fair feature contributions from SHAP.

Preferred reason over other popular approaches:

- **Consistency and Robustness**: Both SHAP and LIME are highly regarded for their reliability in providing explanations that align with human intuition, a critical factor in healthcare applications.
- **Comprehensive Coverage**: SHAP offers global interpretability, while LIME focuses on local explanations. Combining both provides a comprehensive understanding of the model's behavior across different scenarios.
- **Practicality**: PIA is computationally efficient and straightforward to implement, making it a suitable choice for an initial global feature importance analysis.
- **Compatibility with Complex Models**: These methods are model-agnostic, meaning they can be applied to a wide range of ML models, enhancing their versatility.

In comparison to other approaches like LRP (Layer-wise Relevance Propagation) or DeepLIFT (Deep Learning Important FeaTures), which are suitable for deep neural networks, SHAP, LIME, and PIA are better suited for the hybrid ML models used in our study. They offer a good balance between interpretability, computational efficiency, and ease of implementation.

The components of the HXAI-ML framework interact seamlessly to create a cohesive and robust approach to heart disease detection. The process begins with the acquisition and preprocessing of heart disease datasets, where scaling and label mapping standardize the data, ensuring consistency across features. This standardized data is then balanced using hybrid techniques that combine oversampling and undersampling methods, such as RO, TL, SM, and IHT, to mitigate class imbalance. The balanced dataset forms the foundation for training ensemble machine learning models like RFC, DTC, GBC, and XGBC, which are specifically chosen for their ability to capture complex patterns and provide accurate classifications of heart disease stages. Explainable AI tools—SHAP, LIME, and PIA—are then integrated to offer interpretability by quantifying feature contributions and explaining individual and global predictions. These tools create a feedback loop, ensuring that the insights from the XAI components validate and refine the model's decisions. Finally, performance evaluation metrics such as accuracy, precision, and ROC-AUC assess the reliability of the framework, while the interpretable outputs guide clinical decision-making. This interconnection of preprocessing, data balancing, modeling, and XAI ensures a transparent, scalable, and clinically viable solution for heart disease detection.

4. Performance analysis

The performance analysis of our proposed hybrid model was conducted using two widely recognized datasets: the Cleveland Heart Disease Dataset and the Framingham Heart Disease Dataset. These datasets provide a robust foundation for evaluating the effectiveness of our approach across different distributions and feature spaces. Our primary goal was to assess the model's ability to accurately detect and classify heart disease cases, leveraging the hybrid data balancing technique.

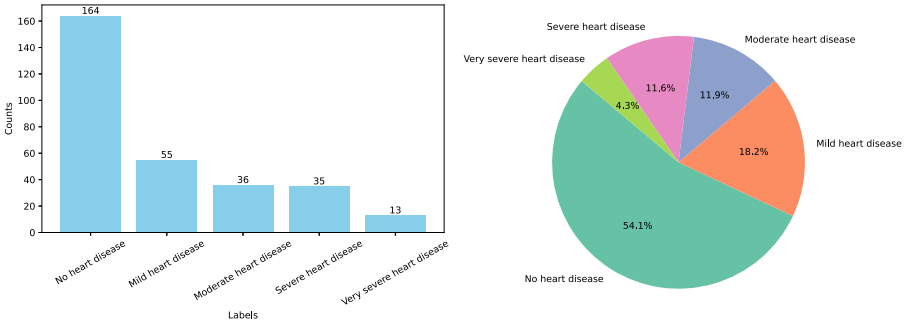
The experiments were carried out by training and testing the model with multiple ML algorithms, followed by a detailed comparison based on various performance metrics. The analysis focused on identifying the best-performing model in terms of predictive accuracy, robustness, and computational efficiency. The results offer insights into the strengths and limitations of each algorithm, paving the way for optimizing cardiovascular disease detection systems.

4.1. Experiment setup

Table 2 provides a comprehensive overview of the datasets, preprocessing steps, ML algorithms, and evaluation metrics used in this study. It also details the hardware specifications and software libraries employed for model training, data manipulation, and evaluation, ensuring a robust and efficient experimental setup for heart disease prediction.

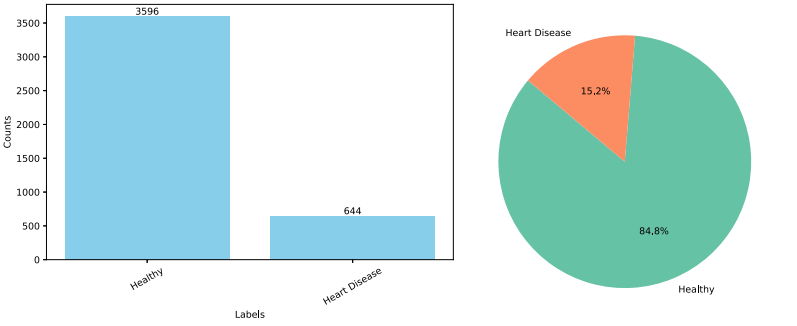
Data collection To develop and evaluate the proposed HXAI-ML model, two publicly available datasets were utilized: the Cleveland Heart Disease Dataset and the Framingham Heart Disease Dataset. These datasets are widely used in cardiovascular research, providing diverse patient populations and essential clinical features for predicting heart disease.

Cleveland heart disease dataset The Cleveland Heart Disease Dataset, sourced from the UCI ML Repository, consists of 303 records with 14 attributes (303, 14). The attributes include both demographic and clinical variables such as age, sex, resting blood pressure, cholesterol levels, fasting blood sugar, electrocardiogram results, and others. The dataset's target variable categorizes patients into five severity classes: 0: No heart disease (164 instances) 1: Mild heart disease (55 instances) 2: Moderate heart disease (36 instances) 3: Severe heart disease (35 instances) 4:



(a) The Count of Each Label (b) The Percentage of Each Label

Fig. 5. Original Data Distribution of Cleveland Dataset.



(a) The Count of Each Label (b) The Percentage of Each Label

Fig. 6. Original Data Distribution of Framingham Dataset.

Very severe heart disease (13 instances) The dataset is often reformulated for binary classification tasks to distinguish between the presence (classes 1–4) and absence (class 0) of heart disease. Fig. 5 illustrates the distribution of the labels in both count and percentage formats, highlighting the imbalanced nature of the dataset.

Framingham heart disease dataset The Framingham Heart Disease Dataset is derived from the long-term Framingham Heart Study and contains 4,240 records with 16 attributes (4240, 16). The features include critical demographic and clinical indicators such as age, gender, smoking status, diabetes, blood pressure, cholesterol levels, body mass index (BMI), and glucose levels. The target variable is binary, identifying whether an individual experienced a cardiovascular event over a 10-year period: 0: Healthy (3,596 instances) 1: Heart Disease (644 instances) This dataset provides a larger and more balanced sample for model training and testing. The label distribution is visualized in Fig. 6, demonstrating the proportion of healthy individuals versus those with heart disease.

Data visualization Figs. 5 and 6 provide detailed visualizations of the original data distributions for both datasets. These insights are critical for understanding label imbalances and informing subsequent data preprocessing and sampling strategies.

Performance metrics To evaluate the performance of our model, we used standard metrics, including Accuracy (Acc), Precision (Pre), Recall (Rec), F1 Score, Cohen's Kappa, Matthews Correlation Coefficient (MCC), ROC-AUC, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Definitions and formulas for these metrics can be found in [61,62]. These metrics collectively provide a holistic view of the model's predictive capabilities and robustness, enabling a detailed comparison of its performance across different datasets and balancing techniques.

4.2. Result analysis

In our experimental analysis, we evaluated the performance of the proposed hybrid model RO+TL+ETC on two benchmark CVD datasets: the Cleveland CVD dataset and the Framingham CVD dataset. The analysis investigates the impact of different ML models and data balancing techniques, including Without Balancing, RO, SM, and RO+TL. These techniques were applied to models such asRF Classifier (RFC), DT Classifier (DTC), GB Classifier (GBC), Extreme GB Classifier (XGBC), and Extra Trees Classifier (ETC), and the results were analyzed based on several performance metrics including accuracy, precision, recall, F1-score, kappa, MCC, ROC-AUC, MAE, MSE, and RMSE.

Cleveland CVD dataset analysis For the Cleveland CVD dataset, we tested various data balancing techniques, observing significant improvements in model performance with the application of RO+TL. As shown in Table 3, the ETC model, when combined with RO+TL, achieved the highest performance metrics, including an accuracy of 98.78%, precision of 98.82%, and an F1-score of 98.77%. Additionally, the model demonstrated high kappa (98.47) and MCC (98.49), which indicate its strong ability to differentiate between positive and negative instances in the dataset. These results highlight that the combination of RO+TL not only balances the dataset but also leverages ML to enhance the generalization of the ETC model, thereby outperforming other techniques and models.

Framingham CVD dataset analysis Similarly, for the Framingham CVD dataset, the RO+TL+ETC hybrid model demonstrated superior performance across multiple metrics. As illustrated in Table 4, this model achieved an accuracy of 96.78%, precision of 97.12%, and an F1-score of 96.87%. These results are significantly better than those of other combinations, showcasing the robustness of the proposed approach. The improved performance can be attributed to the synergistic effect of data balancing with RO and the leveraging of ML to adapt the model to the

Table 3
Performance Analysis of ML Models on Cleveland CVD Dataset.

Exp. No.	Data Balancing Technique	Model	Accuracy	Precision	Recall	F1score	Kappa	MCC	ROC-AUC	MAE	MSE	RMSE
		RFC	63.93	63.33	63.93	60.71	41.45	42.6	82.48	52.46	91.8	95.81
		DTC	52.46	56.83	52.46	51.5	27.32	27.93	65.13	85.25	183.61	135.5
1	Without	GBC	57.38	56.82	57.38	55.74	31.76	32.24	84.82	60.66	100	100
		XGBC	62.3	60.36	62.3	60.96	39.81	40.05	83.99	54.1	93.44	96.67
		ETC	60.66	57.97	60.66	56.14	33.64	35.11	85.26	60.66	116.39	107.89
		RFC	94.51	94.78	94.51	94.54	93.11	93.17	99.82	6.71	10.37	32.2
		DTC	92.68	92.9	92.68	92.52	90.78	90.92	95.37	15.85	43.9	66.26
2	RO	GBC	94.51	94.72	94.51	94.43	93.1	93.18	99.61	9.15	21.34	46.2
		XGBC	92.68	92.78	92.68	92.59	90.8	90.87	99.69	9.15	14.02	37.45
		ETC	95.12	95.36	95.12	95.14	93.88	93.93	99.76	6.1	8.54	29.22
		RFC	87.8	88.33	87.8	87.78	84.69	84.83	98.56	17.07	28.05	52.96
		DTC	72.56	73.52	72.56	72.72	65.59	65.76	82.99	45.12	92.68	96.27
3	SM	GBC	81.71	82.68	81.71	81.67	77.01	77.28	96.84	25	39.63	62.96
		XGBC	88.41	88.59	88.41	88.35	85.43	85.5	98.38	17.07	30.49	55.22
		ETC	94.51	94.58	94.51	94.43	93.09	93.15	99.42	9.15	18.9	43.48
		RFC	96.95	96.94	96.95	96.94	96.18	96.18	99.8	3.66	4.88	22.09
		DTC	94.51	94.81	94.51	94.49	93.13	93.22	96.65	6.1	7.32	27.05
4	**RO+TL**	GBC	96.34	96.31	96.34	96.3	95.41	95.43	98.93	5.49	10.37	32.2
		XGBC	97.56	97.56	97.56	97.55	96.94	96.94	99.55	3.66	7.32	27.05
		ETC	**98.78**	**98.82**	**98.78**	**98.77**	**98.47**	**98.49**	**99.97**	**2.44**	**6.1**	**24.69**
		RFC	91.46	91.61	91.46	91.36	89.26	89.35	98.63	10.98	17.07	41.32
		DTC	71.34	72.19	71.34	71.25	63.98	64.21	82.13	45.12	89.02	94.35
5	SM+TL	GBC	81.71	82.68	81.71	81.67	77.01	77.28	96.85	25	39.63	62.96
		XGBC	88.41	88.59	88.41	88.35	85.43	85.5	98.38	17.07	30.49	55.22
		ETC	90.24	90.33	90.24	90.11	87.72	87.81	99.37	13.41	23.17	48.14
		RFC	93.9	94.1	93.9	93.88	92.31	92.37	98.99	7.32	9.76	31.23
		DTC	91.46	91.45	91.46	91.23	89.24	89.37	94.53	14.63	34.15	58.43
6	RO+IHT	GBC	91.46	91.79	91.46	91.35	89.24	89.39	97.98	9.76	12.2	34.92
		XGBC	91.46	91.62	91.46	91.34	89.24	89.35	98.38	10.37	15.24	39.04
		ETC	92.68	92.85	92.68	92.61	90.77	90.85	99.47	8.54	10.98	33.13
		RFC	89.02	89.1	89.02	88.89	86.21	86.3	98.74	14.63	21.95	46.85
		DTC	78.05	78.54	78.05	77.93	72.42	72.57	86.33	31.1	51.83	71.99
7	SM+IHT	GBC	83.54	83.39	83.54	83.25	79.31	79.41	96.56	25.61	48.78	69.84
		XGBC	87.2	87.32	87.2	86.96	83.91	84.04	97.65	21.34	43.29	65.8
		ETC	90.85	91.08	90.85	90.85	88.51	88.57	98.89	14.02	27.44	52.38

specific characteristics of the dataset. The high ROC-AUC and low error metrics further underscore the effectiveness of RO+TL in optimizing model performance for both Cleveland and Framingham CVD datasets.

Comparative performance and significance The overall comparative analysis of the models on both datasets reveals that the hybrid RO+TL+ETC model consistently outperforms other methods. While the base models, such as RFC and GBC, exhibited satisfactory results with data balancing techniques like RO and SM, the incorporation of ML provided a notable boost in model accuracy and stability. The enhanced performance of RO+TL+ETC, particularly in terms of kappa, MCC, and F1-score, confirms that this hybrid approach is highly effective in addressing the challenges posed by imbalanced datasets and in improving the predictive power of ML models for cardiovascular disease prediction. This highlights the importance of combining data balancing strategies with advanced learning techniques to achieve optimal performance in medical prediction tasks.

The best-performing model analysis In the experiments conducted on the Cleveland and Framingham CVD datasets, the best-performing model across all scenarios was achieved with the combination of RO and the ETC with the application of TL for data balancing. For the Cleveland dataset (Table 5), this model achieved an impressive accuracy of 98.78%, along with outstanding metrics across precision, recall, F1-score, and ROC-AUC. The model's robustness is further demonstrated by its low MAE, MSE, and RMSE, indicating efficient error minimization. Similarly, for the Framingham dataset (Table 6), the model utilizing RO and TL resulted in a remarkable accuracy of 99.24%, with precision and recall values also reaching peak performance. This combination of data balancing techniques and model architecture notably enhanced the prediction capabilities, outperforming other configurations and highlighting the effectiveness of RO+TL+ETC in handling complex cardiovascular datasets.

Classification report analysis The classification report analysis for both Cleveland and Framingham CVD datasets demonstrates the impact of using Random Oversampling combined with Tomek Links (RO+TL) for data balancing, as well as different ML models. In the Cleveland CVD dataset (Table 7), the ETC achieved exceptional performance across all classes, with 100% precision, recall, and F1-score for severe and very severe heart disease, while maintaining a high performance for other classes like mild and moderate heart disease. The DTC and GBC also performed well, with f1-scores of around 98 for moderate and severe heart disease. On the Framingham dataset (Table 8), ETC again outperformed other models, with remarkable precision and recall of 99.59% and 98.91% for healthy individuals, and 98.88% precision and 99.58% recall for heart disease cases. The RFC also demonstrated strong performance, with an F1-score of 97.64 for heart disease detection. Overall, the use of RO+TL significantly enhanced model performance, particularly in distinguishing between different levels of heart disease severity, with ETC emerging as the most reliable model across both datasets. The high precision and recall values across both healthy and heart disease classes validate the robustness of these models in classifying cardiovascular conditions.

Table 4
Performance Analysis of ML Models on Framingham CVD Dataset.

Exp. No.	Data Balancing Technique	Model	Accuracy	Precision	Recall	F1score	Kappa	MCC	ROC-AUC	MAE	MSE	RMSE
		RFC	84.32	78.88	84.32	78.48	5.64	10.53	63.76	15.68	15.68	39.6
		DTC	76.65	76.51	76.65	76.58	10.62	10.62	55.28	23.35	23.35	48.32
1	Without	GBC	84.55	80.18	84.55	79.16	8.84	14.75	66.14	15.45	15.45	39.3
		XGBC	83.25	77.87	83.25	79.14	10.22	12.71	62.23	16.75	16.75	40.92
		ETC	84.67	80.78	84.67	79.06	8.22	14.86	64.92	15.33	15.33	39.15
		RFC	97.57	97.65	97.57	97.57	95.14	95.21	99.8	2.43	2.43	15.6
		DTC	90.34	91.75	90.34	90.28	80.74	82.11	90.52	9.66	9.66	31.08
2	RO	GBC	73.59	73.95	73.59	73.54	47.28	47.58	82.13	26.41	26.41	51.39
		XGBC	92.15	92.85	92.15	92.13	84.33	85.01	97.84	7.85	7.85	28.02
		ETC	98.89	98.89	98.89	98.89	97.78	97.78	99.79	1.11	1.11	10.54
		RFC	89.99	90.01	89.99	89.99	79.98	80	96.21	10.01	10.01	31.63
		DTC	79.57	79.58	79.57	79.57	59.13	59.14	79.56	20.43	20.43	45.2
3	SM	GBC	81.45	81.45	81.45	81.44	62.89	62.9	89.61	18.55	18.55	43.07
		XGBC	89.16	89.21	89.16	89.16	78.32	78.37	95.39	10.84	10.84	32.93
		ETC	90.41	90.53	90.41	90.4	80.81	80.94	97.39	9.59	9.59	30.97
		RFC	97.64	97.71	97.64	97.64	95.27	95.35	99.68	2.36	2.36	15.38
		DTC	92.14	93.11	92.14	92.11	84.32	85.26	92.26	7.86	7.86	28.03
4	**RO+TL**	GBC	72.67	72.77	72.67	72.66	45.38	45.45	80.72	27.33	27.33	52.28
		XGBC	92.14	92.67	92.14	92.13	84.31	84.82	97.93	7.86	7.86	28.03
		ETC	**99.24**	**99.24**	**99.24**	**99.24**	**98.47**	**98.47**	**99.67**	**0.76**	**0.76**	**8.75**
		RFC	89.16	89.25	89.16	89.15	78.31	78.41	96.04	10.84	10.84	32.93
		DTC	79.71	79.71	79.71	79.71	59.41	59.42	79.7	20.29	20.29	45.05
5	SM+TL	GBC	81.45	81.45	81.45	81.44	62.89	62.9	89.58	18.55	18.55	43.07
		XGBC	89.16	89.21	89.16	89.16	78.32	78.37	95.39	10.84	10.84	32.93
		ETC	91.04	91.18	91.04	91.03	82.06	82.21	97.58	8.96	8.96	29.94
		RFC	97.15	97.25	97.15	97.15	94.3	94.4	99.55	2.85	2.85	16.88
		DTC	91.1	92.28	91.1	91.05	82.22	83.39	91.13	8.9	8.9	29.82
6	RO+IHT	GBC	70.95	71.1	70.95	70.91	41.92	42.06	77.81	29.05	29.05	53.9
		XGBC	90.97	91.59	90.97	90.93	81.94	82.56	96.54	9.03	9.03	30.06
		ETC	98.68	98.69	98.68	98.68	97.36	97.37	99.79	1.32	1.32	11.49
		RFC	88.81	88.83	88.81	88.81	77.6	77.61	95.85	11.19	11.19	33.45
		DTC	82.42	82.42	82.42	82.4	64.73	64.76	82.32	17.58	17.58	41.93
7	SM+IHT	GBC	81.93	81.99	81.93	81.94	63.85	63.89	90.97	18.07	18.07	42.51
		XGBC	90.13	90.24	90.13	90.14	80.27	80.36	95.2	9.87	9.87	31.41
		ETC	91.94	91.94	91.94	91.94	83.85	83.85	97.79	8.06	8.06	28.39

Table 5
The best-performing model from each experiment on Cleveland CVD Dataset.

Exp. No.	Data Balancing Technique	Model	Accuracy	Precision	Recall	F1score	Kappa	MCC	ROC-AUC	MAE	MSE	RMSE
1	Without	RFC	63.93	63.33	63.93	60.71	41.45	42.6	82.48	52.46	91.8	95.81
2	RO	ETC	95.12	95.36	95.12	95.14	93.88	93.93	99.76	6.1	8.54	29.22
3	SM	ETC	94.51	94.58	94.51	94.43	93.09	93.15	99.42	9.15	18.9	43.48
4	**RO+TL**	**ETC**	**98.78**	**98.82**	**98.78**	**98.77**	**98.47**	**98.49**	**99.97**	**2.44**	**6.1**	**24.69**
5	SM+TL	RFC	91.46	91.61	91.46	91.36	89.26	89.35	98.63	10.98	17.07	41.32
6	RO+IHT	RFC	93.9	94.1	93.9	93.88	92.31	92.37	98.99	7.32	9.76	31.23
7	SM+IHT	ETC	90.85	91.08	90.85	90.85	88.51	88.57	98.89	14.02	27.44	52.38

Table 6
The best-performing model from each experiment on Framingham CVD Dataset.

Exp. No.	Data Balancing Technique	Model	Accuracy	Precision	Recall	F1score	Kappa	MCC	ROC-AUC	MAE	MSE	RMSE
1	Without	ETC	84.67	80.78	84.67	79.06	8.22	14.86	64.92	15.33	15.33	39.15
2	RO	RFC	97.57	97.65	97.57	97.57	95.14	95.21	99.8	2.43	2.43	15.6
3	SM	ETC	90.41	90.53	90.41	90.4	80.81	80.94	97.39	9.59	9.59	30.97
4	**RO+TL**	**ETC**	**99.24**	**99.24**	**99.24**	**99.24**	**98.47**	**98.47**	**99.67**	**0.76**	**0.76**	**8.75**
5	SM+TL	ETC	91.04	91.18	91.04	91.03	82.06	82.21	97.58	8.96	8.96	29.94
6	RO+IHT	ETC	98.68	98.69	98.68	98.68	97.36	97.37	99.79	1.32	1.32	11.49
7	SM+IHT	ETC	91.94	91.94	91.94	91.94	83.85	83.85	97.79	8.06	8.06	28.39

Table 7
Classification Reports of ML Model Using RO+TL on Cleveland CVD Dataset.

Exp. No.	Data Balancing Technique	Model	Class	Precision	Recall	F1-score
Proposed Experiment	Random Oversampling + TomekLink (RO+TL)	DTC	No heart disease	93.75	81.08	86.96
			Mild heart disease	80.65	92.59	86.21
			Moderate heart disease	96.88	100	98.41
			Severe heart disease	100	100	100
			Very severe heart disease	100	100	100
		ETC	**No heart disease**	**100**	**94.59**	**97.22**
			Mild heart disease	**96.43**	**100**	**98.18**
			Moderate heart disease	**100**	**100**	**100**
			Severe heart disease	**96.97**	**100**	**98.46**
			Very severe heart disease	**100**	**100**	**100**
		GBC	No heart disease	94.29	89.19	91.67
			Mild heart disease	92.59	92.59	92.59
			Moderate heart disease	96.88	100	98.41
			Severe heart disease	96.97	100	98.46
			Very severe heart disease	100	100	100
		RFC	No heart disease	94.44	91.89	93.15
			Mild heart disease	92.59	92.59	92.59
			Moderate heart disease	96.88	100	98.41
			Severe heart disease	100	100	100
			Very severe heart disease	100	100	100
		XGBC	No heart disease	94.59	94.59	94.59
			Mild heart disease	96.15	92.59	94.34
			Moderate heart disease	100	100	100
			Severe heart disease	96.97	100	98.46
			Very severe heart disease	100	100	100

Table 8
Classification Reports of ML Model Using RO+TL on Framingham CVD Dataset.

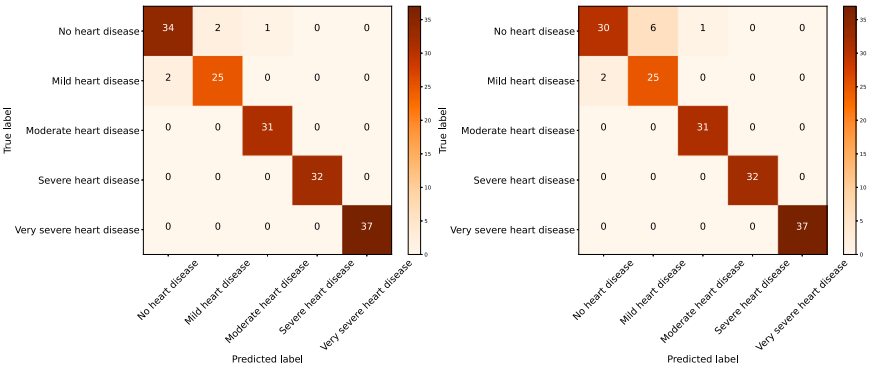
Exp. No.	Data Balancing Technique	Model	Class	Precision	Recall	F1-score
Proposed Experiment	RO+TL	DTC	Healthy	99.52	84.95	91.66
			Heart Disease	86.49	99.58	92.57
		ETC	**Healthy**	**99.59**	**98.91**	**99.25**
			Heart Disease	**98.88**	**99.58**	**99.22**
		GBC	Healthy	74.49	70.31	72.34
			Heart Disease	70.99	75.11	72.99
		RFC	Healthy	99.57	95.76	97.63
			Heart Disease	95.78	99.58	97.64
		XGBC	Healthy	97.39	86.87	91.83
			Heart Disease	87.79	97.6	92.43

Confusion matrix analysis The confusion matrix analysis for both the Cleveland and Framingham CVD datasets highlights the superior performance of the ETC compared to other ML models. On both datasets, ETC consistently demonstrates the highest accuracy, with a significantly lower number of misclassifications. In the confusion matrices (Fig. 7) for the Cleveland CVD dataset, ETC shows the best performance with 723 true positives for heart disease and 704 true positives for healthy individuals. The number of misclassifications is minimal, with only a few false positives and false negatives, making it the top-performing model. On the other hand, RFC performs well with 700 true positives for healthy and 704 true positives for heart disease, although there are slightly more misclassifications compared to ETC. DTC has more misclassifications, especially with 110 false positives for healthy individuals. GBC (GB Classifier) and XGBC also demonstrate good performance, but there are higher rates of false positives for the healthy class, especially with GBC.

The confusion matrices (Fig. 8) for the Framingham CVD dataset show similar trends, with ETC again being the best-performing model. It correctly identifies 723 true positives for healthy individuals and 704 true positives for heart disease, with very few misclassifications. In comparison, RFC performs excellently with minimal misclassifications, but it still falls short of ETC's accuracy. DTC shows a similar trend, with good classification of heart disease (704 true positives), but with a higher number of false positives (110). GBC and XGBC also perform well but have slightly higher rates of misclassification, particularly with healthy individuals being misclassified as heart disease.

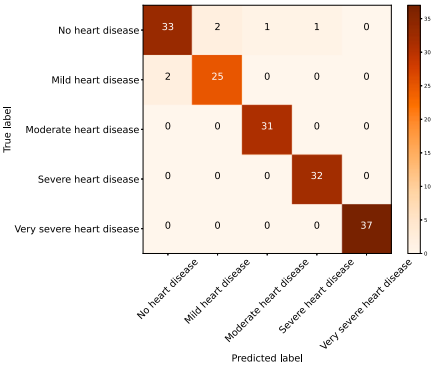
Both datasets show that ETC is the best-performing model across the board, with the highest number of true positives and the lowest misclassifications in both Healthy and Heart Disease categories. The RO+TL framework significantly enhances the classification performance of the ML models, with ETC consistently outperforming all other models (RFC, DTC, GBC, XGBC) on both the Cleveland and Framingham CVD datasets. This demonstrates the effectiveness of ETC in accurately predicting cardiovascular disease outcomes, making it the most reliable model for this task.

Key observations

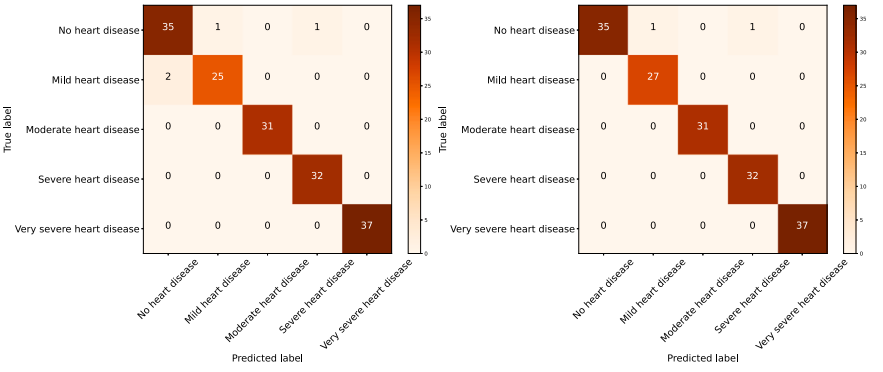


(a) RFC

(b) DTC



(c) GBC



(d) XGBC

(e) ETC

Fig. 7. Confusion Matrix of ML Model Using RO+TL on Cleveland CVD Dataset.

- Impact of Balancing Techniques: Techniques such as RO+TL significantly enhanced the models' ability to handle imbalanced datasets, as evidenced by substantial improvements in recall, F1-score, and ROC-AUC across both datasets.
- Superior Performance of ETC: The Extra Trees Classifier consistently outperformed other models, especially when combined with RO+TL, highlighting its ability to capture complex data patterns effectively.
- Generalization Across Datasets: The hybrid approach demonstrated robust performance across both Cleveland and Framingham datasets, affirming its scalability and adaptability for CVD prediction tasks.

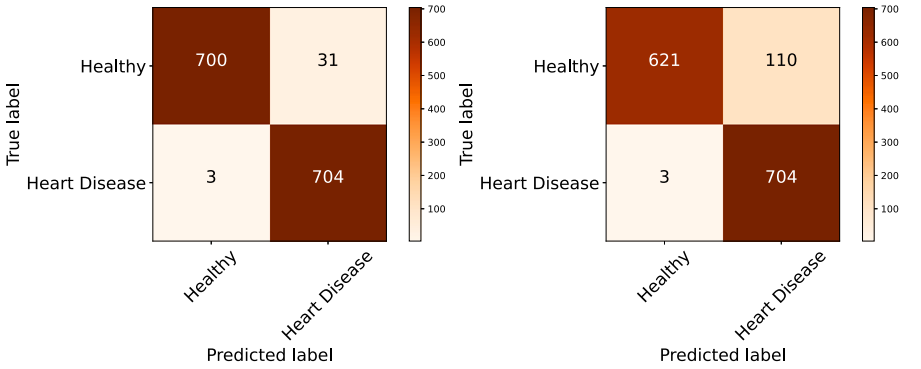
4.3. Explainable AI (XAI) analysis

In this subsection, we present an in-depth XAI analysis for our proposed hybrid ML model, which utilizes RO, TL, and ETC for heart disease detection and classification. The XAI framework enables a transparent understanding of the model's predictions by visualizing the contributions of individual features. This analysis enhances the interpretability of the hybrid model and ensures its practical applicability in clinical decision-making.

4.3.1. XAI analysis for the Cleveland CVD dataset

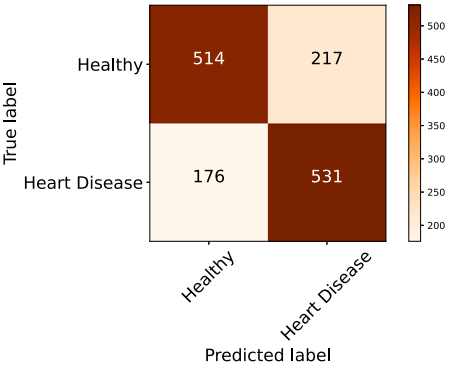
The comprehensive XAI analysis for our proposed hybrid ML model, which incorporates RO, TL, and ETC for CVD detection using the Cleveland dataset. The XAI framework ensures transparency by highlighting the most influential features and explaining their roles in both global and local predictions.

Feature importance using permutation method Feature Importance analysis (Fig. 9a) employs the permutation method (PIA) to evaluate feature importance based on the impact of each feature on model performance. Features like exang, ca, restecg, and thal are identified as the most in-

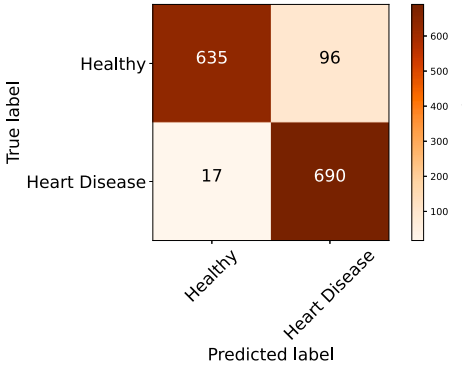


(a) RFC

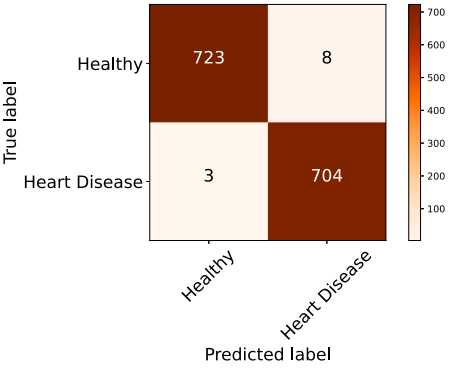
(b) DTC



(c) GBC



(d) XGBC



(e) ETC

Fig. 8. Confusion Matrix of ML Model Using RO+TL on Framingham CVD Dataset.

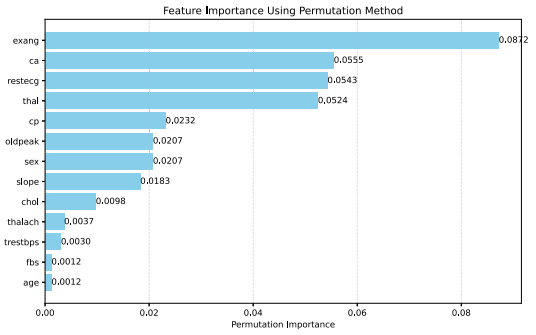
fluential variables in predicting heart disease. Their high permutation importance scores demonstrate their significant role in shaping the model's outcomes. This visualization confirms the robustness of the hybrid ML model in leveraging critical features for accurate predictions.

SHAP (Shapley additive exPlanations) analysis SHAP analysis (Fig. 9b) provides a detailed global explanation of the model's behavior by measuring the contribution of each feature to the model output. The SHAP plot indicates the positive and negative impacts of features like ca, thal, and exang on the predicted probability of heart disease. High feature values (e.g., ca and thal) are strongly associated with an increased likelihood of heart disease, as indicated by the red gradient. This ensures that the model's predictions align with clinical relevance.

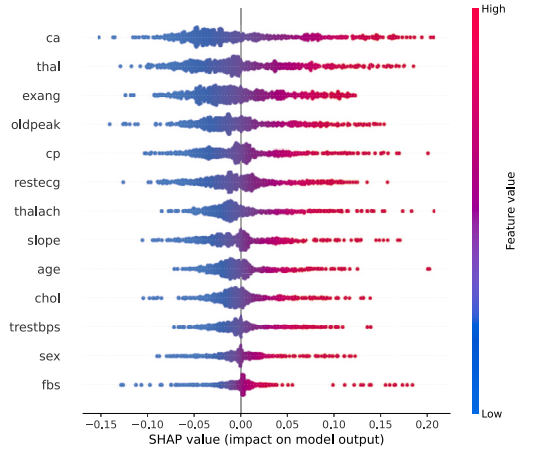
Local interpretation using LIME The Local Interpretable Model-Agnostic Explanations (LIME) framework (Fig. 9c) provides an individual-level explanation for heart disease prediction. For a specific instance from the Cleveland CVD dataset, LIME highlights the most critical features contributing to the classification. Features such as g, s, and a significantly support the predicted class, while others like xl and il negatively influence the prediction. This instance-level interpretation offers clinicians actionable insights into why a specific prediction was made.

The SHAP analysis visualizes the feature contributions for different classes of heart disease severity predicted by our proposed model on the Cleveland CVD dataset (Fig. 10). Each subfigure highlights the role of critical features like age, sex, ca, and thal in distinguishing between no, mild, moderate, severe, and very severe heart disease cases, ensuring transparency and interpretability across all prediction classes.

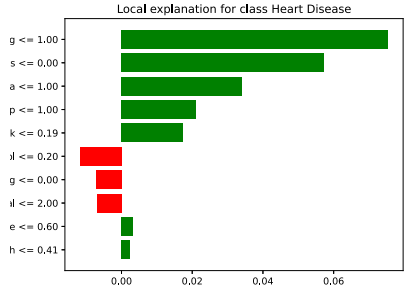
Overall insights on Cleveland dataset The combination of PIA, SHAP, and LIME analyses validates the effectiveness of the hybrid ML model in detecting and classifying heart disease. These XAI methods reveal that the model accurately identifies key predictive features, while also providing transparent explanations for its predictions. Such interpretability



(a) Feature Importance



(b) SHAP



(c) LIME

Fig. 9. Analysis of XAI of our proposed Model on Cleveland CVD Dataset.

enhances trustworthiness and facilitates the integration of the model into real-world medical applications.

4.3.2. XAI analysis for the framingham CVD dataset

The comprehensive XAI analysis for our proposed hybrid ML model, which incorporates RO, TL, and ETC for CVD detection using the Framingham dataset. The XAI framework ensures transparency by highlighting the most influential features and explaining their roles in both global and local predictions.

Feature importance using permutation method The permutation importance analysis (PIA) (Fig. 11a) ranks features based on their impact on model performance. Key features such as age (0.1461), male (0.1369), and prevalentHyp (0.1086) are identified as the most critical predictors of CVD. Features like sysBP and diaBP also contribute significantly, reflecting the importance of blood pressure-related factors in CVD detection. The results validate the ability of the hybrid ML model to prioritize clinically relevant features, demonstrating its robustness in predictive analysis.

SHAP (Shapley additive exPlanations) analysis SHAP analysis (Fig. 11b) provides a detailed understanding of the contribution of individual features to the model's predictions. The SHAP value plot illustrates the influence of features such as age, male, and prevalentHyp on model output. Features with higher values, such as age and sysBP, positively impact the likelihood of predicting CVD, while features like education and BMI exhibit varied effects based on their values. The gradient from low (blue) to high (red) feature values offers a clear interpretation of how individual features shape the model's decisions, enhancing its interpretability.

Local explanation using LIME The LIME framework (Fig. 11c) provides localized insights into individual predictions. For a specific instance, LIME highlights the dominant factors influencing the classification of CVD. Features such as prevalentHyp, prevalentStroke, and male play a critical role in the prediction, while features like BPMeds and diabetes contribute negatively in this example. This local explanation aligns with the model's global interpretability findings, reinforcing the trustworthiness of the hybrid model in real-world clinical contexts.

The SHAP analysis illustrates the feature contributions for two classes, Healthy and Heart Disease, as predicted by our proposed model on the Framingham CVD dataset (Fig. 12). Subfigure (a) highlights how features like age, male, and prevalentHyp influence predictions for healthy individuals, while subfigure (b) demonstrates their impact on identifying heart disease cases. This analysis ensures interpretability by showcasing the role of key features in distinguishing between the two classes.

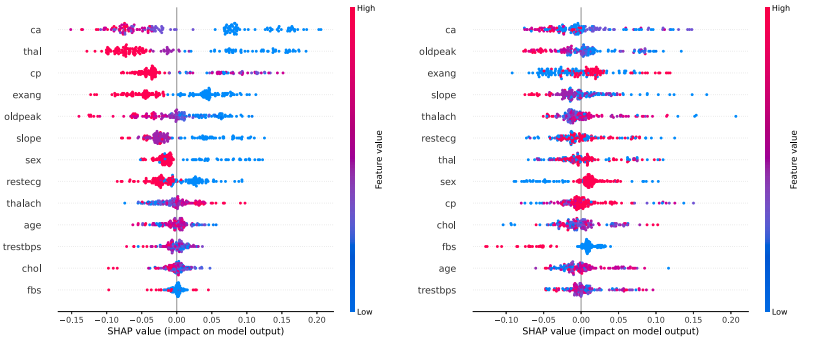
Overall insights on framingham dataset The XAI analysis across PIA, SHAP, and LIME confirms the effectiveness and reliability of the proposed hybrid ML model. The ability to explain predictions at both global and local levels ensures transparency, a crucial requirement for deploying ML models in medical decision-making. By consistently identifying and prioritizing relevant features such as age, blood pressure metrics, and prevalentHyp, the model demonstrates strong alignment with established cardiovascular risk factors, making it a valuable tool for clinicians.

4.3.3. Comprehensive XAI analysis: exploring global and multiple instance explanations

To position our hybrid ML model as a truly explainable tool, we extend the XAI analysis beyond individual instance-level explanations. By analyzing multiple instances and identifying global patterns in feature importance, we gain deeper insights into the model's decision-making process. This broader analysis enhances the model's interpretability and trustworthiness, ensuring that it can be confidently applied in real-world scenarios.

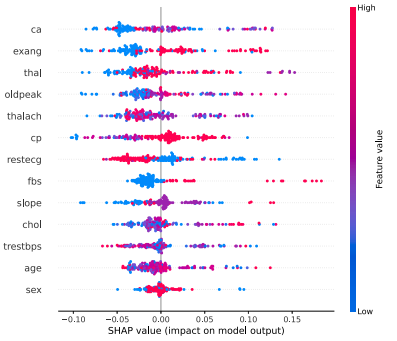
Global feature importance across the dataset One critical aspect of explainable AI is the ability to understand how features influence predictions across the entire dataset, rather than on a single instance. To achieve this, we leverage the global feature importance analysis, which evaluates the impact of each feature on the model's predictions across all instances in the dataset. This approach ensures that the model is not only interpretable at the individual level but also provides a holistic view of which features are consistently important across various data points.

Figs. 9a and 11a provide a global view of feature importance for the Cleveland and Framingham datasets, respectively. In these analyses, we observe that features such as age, sex, blood pressure, and specific health indicators (like 'ca' and 'thal' for Cleveland, and 'sysBP' and 'prevalentHyp' for Framingham) are consistently ranked as the most important features across the datasets. This global feature ranking solidifies the

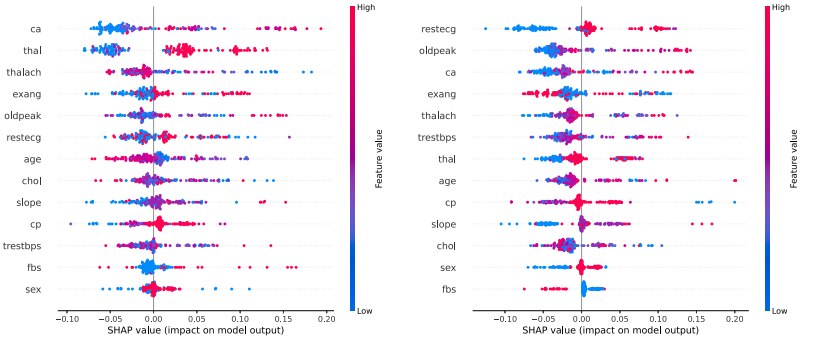


(a) SHAP for No heat disease

(b) SHAP for Mild heart disease



(c) SHAP for Moderate heart disease



(d) SHAP for Severe heart disease

(e) SHAP for Very severe heart disease

Fig. 10. Analysis of SHAP for various classes of our proposed Model on Cleveland CVD Dataset.

model's reliance on clinically relevant predictors, which enhances its practical applicability in clinical settings.

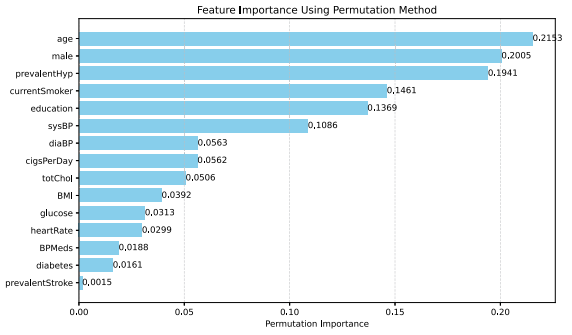
Trend analysis of explanations across instances A crucial step in understanding model behavior is analyzing trends and patterns in feature importance over multiple instances. This trend analysis helps identify whether the model exhibits consistent behavior across different demographic groups, severity levels of heart disease, or varying input conditions. For example, we can examine whether age and blood pressure metrics consistently have a significant impact on predictions, regardless of the specific instance or class.

By aggregating SHAP values across all instances in the Cleveland and Framingham datasets, we observe patterns in how features like age, sex, and blood pressure-related variables affect the model's predictions across different instances. This aggregated view (Fig. 10 and 12) provides deeper insights into the model's decision-making process, revealing trends in how the model weighs these features in detecting and classifying heart disease. Such trends further validate the model's reliance on established cardiovascular risk factors and increase its credibility as an explainable tool.

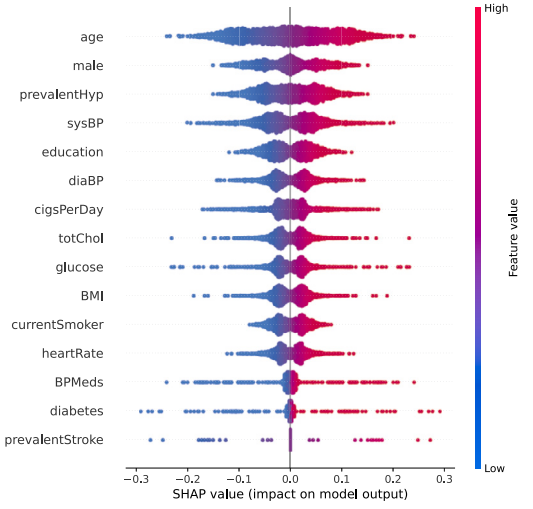
Global and local patterns in LIME explanations While LIME has been traditionally used for local explanations, it can also be extended to analyze global trends by aggregating local explanations over multiple instances. This aggregated view (Figs. 9c and 11c) allows us to identify how local feature importance varies across instances in the dataset, highlighting whether certain features are consistently influential across multiple predictions.

For example, in the Cleveland dataset, features such as 'thal' and 'ca' are shown to be consistently important for the classification of severe and very severe heart disease cases, while features like 'restecg' and 'exang' play a more significant role in predicting mild cases. This analysis reinforces the model's ability to adapt to different levels of heart disease severity, ensuring that its predictions remain relevant and interpretable at both local and global scales.

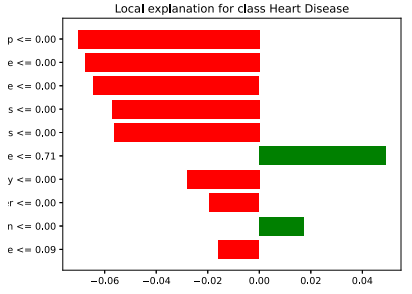
Overall insights from comprehensive XAI analysis The comprehensive XAI analysis emphasizes the importance of moving beyond single-instance explanations. By incorporating global feature importance, trend analysis, and aggregated local explanations, we provide a more thorough understanding of the model's behavior. This in-depth analysis not only



(a) Feature Importance



(b) SHAP



(c) LIME

Fig. 11. Analysis of XAI of our proposed Model on Framingham CVD Dataset.

validates the model's reliance on clinically relevant features but also ensures that the model's decisions are interpretable across various instances and demographic groups. These insights strengthen the model's transparency, building trust and confidence in its deployment within real-world clinical environments.

4.4. Complexity analysis

To evaluate the computational efficiency of the proposed approach, we conducted a detailed complexity analysis by measuring the training time, prediction time, and total time required by various ML models when integrated with the RO+TL data balancing technique. The results are presented in Table 9 and Table 10 for the Cleveland and Framingham CVD datasets, respectively.

For the Cleveland dataset (Table 9), the models demonstrated varied computational demands. The RFC and Extremely Randomized Trees Classifier (ETC) exhibited moderate training and prediction times, with total times of 0.349 seconds and 0.320 seconds, respectively. The DTC had the lowest total time of 0.017 seconds, making it the most computationally efficient. GBC and XGBoost Classifier (XGBC), while more computationally intensive, still delivered manageable total times of 1.261 seconds and 0.322 seconds, respectively. For the Framingham dataset (Table 10), the computational overhead increased due to the larger dataset size. RFC and ETC required 1.340 seconds and 0.913 seconds, respectively, while DTC remained the fastest with a total time of 0.048 seconds. GBC and XGBC maintained a balance between performance and computational demand, with total times of 1.350 seconds and 0.264 seconds, respectively.

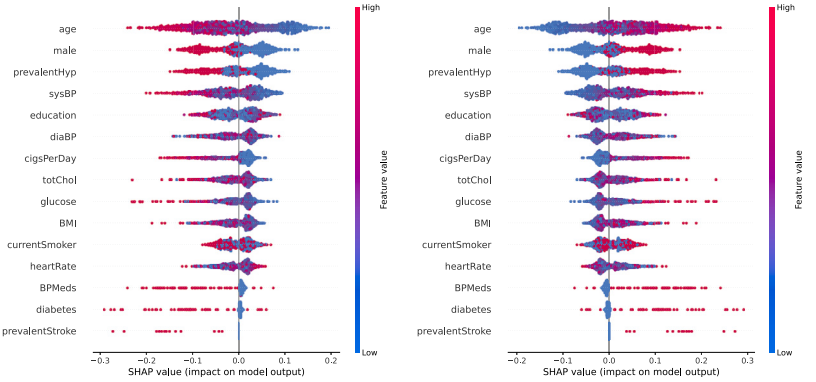
The complexity analysis highlights the proposed approach's efficiency, demonstrating that the RO+TL data balancing technique does not introduce significant computational overhead, even when applied to larger datasets. Among the evaluated models, ETC emerged as a robust candidate, balancing predictive performance with computational efficiency.

5. Discussion

In this study, we have proposed a novel heart disease prediction model, HXAI-ML, and performed an extensive evaluation on two widely used heart disease datasets: the Cleveland and Framingham datasets. The goal of this comparison was to highlight the performance and robustness of HXAI-ML in predicting heart disease while also addressing class imbalance using advanced data balancing techniques such as Random Oversampling (RO) and Tomek Links (TL). The results of this study are benchmarked against several state-of-the-art models from recent literature, showcasing HXAI-ML's superiority in both prediction accuracy and interpretability, thanks to the integration of powerful ML algorithms and Explainable AI (XAI) techniques like SHAP, LIME, and PIA. In the following sections, we provide detailed discussions on the comparative performance of HXAI-ML on both datasets.

On the Cleveland dataset (Table 11), HXAI-ML outperformed several top-performing models, achieving an accuracy of 98.78%. Notably, Mienye et al.'s XGBoost model, which used SHAP for explainability, achieved 98.4%, while Kayalvizhi et al.'s CNN-BiLSTM model reached 95.3%, and Elsevimy et al.'s QPSO-SVM achieved 96.31%. Other models, such as Nazari et al.'s Ensemble model (88.05%) and Ranade et al.'s LR + SFS (84.4%), demonstrated significantly lower accuracy, highlighting the robustness of HXAI-ML. The superiority of HXAI-ML can be attributed to its advanced data balancing methods, particularly the combination of RO and TL, which helped mitigate class imbalance and improve model performance. Additionally, the use of the Extra Trees Classifier (ETC) along with Explainable AI techniques, such as SHAP, LIME, and PIA, ensured both high accuracy and model transparency. These results position HXAI-ML as a highly competitive model for heart disease detection on the Cleveland dataset, offering not just improved prediction accuracy but also the interpretability crucial for clinical applications.

Turning to the Framingham dataset (Table 12), HXAI-ML achieved an outstanding accuracy of 99.24%, setting a new benchmark for heart disease prediction on this dataset. This performance surpassed several other prominent models in the literature. For instance, Omotehinwa et al.'s LGBM model achieved 98.82%, Rahim et al.'s MaLCaDD achieved 99.1%, and Mienye et al.'s Ensemble model reached 91%. The excellent performance of HXAI-ML can be attributed to the synergistic effect of robust data balancing techniques (RO+TL) and the powerful ETC algorithm. Additionally, the incorporation of SHAP, LIME, and PIA for model explainability provides not only improved accuracy but also transparency, which is crucial in healthcare contexts. The performance of HXAI-ML stands out particularly when compared to models such as



(a) SHAP for Healthy (b) SHAP for heart disease

Fig. 12. Analysis of SHAP for various classes of our proposed Model on Framingham CVD Dataset.

Table 9
Complexity Analysis of ML Models on Cleveland CVD Dataset.

Data Balancing Technique	Model Name	Training Time (s)	Prediction Time (s)	Total Time (s)
RO+TL	RFC	0.318	0.031	0.349
	DTC	0.016	0.001	0.017
	GBC	1.260	0.001	1.261
	XGBC	0.306	0.016	0.322
	ETC	**0.289**	**0.031**	**0.320**

Table 10
Complexity Analysis of ML Models on Framingham CVD Dataset.

Data Balancing Technique	Model Name	Training Time (s)	Prediction Time (s)	Total Time (s)
RO+TL	RFC	1.277	0.063	1.340
	DTC	0.047	0.001	0.048
	GBC	1.334	0.016	1.350
	XGBC	0.248	0.016	0.264
	ETC	**0.85**	**0.062**	**0.913**

Table 11
Comparison Analysis of our Proposed (HXAI-ML) Model with existing works on Cleveland Dataset.

SI. No.	Author	Dataset	Data Balancing	Model	Accuracy	XAI
1	Mienye et al. [27]	Cleveland	X	XGBoost	98.4%	SHAP
2	Kayalvizhi et al. [41]	Cleveland	X	CNN-BiLSTM	95.3%	X
3	Nazari et al. [34]	Cleveland	X	Ensemble	88.05%	X
4	Shokouhifar et al. [29]	Cleveland	X	EHMFFL	91.8%	X
5	Elsevimy et al. [33]	Cleveland	X	QPSO-SVM	96.31%	X
6	Ranade et al. [28]	Cleveland	X	LR + SFS	84.4%	X
7	Chandrasekhar et al. [36]	Cleveland	X	Soft Voting	93.44%	X
8	Anderies et al. [37]	Cleveland	X	SVM	85%	X
9	Srinivas et al. [38]	Cleveland	X	XGBoost	94.7%	X
10	Chhillar et al. [39]	Cleveland	X	SVM	84.15%	X
11	Akella and Akella [40]	Cleveland	X	NN	93%	X
12	Sivakami et al. [30]	Cleveland	X	DT + LR	96%	X
13	Perumal et al. [32]	Cleveland	X	PCA+LR	87%	X
14	Mienye et al. [35]	Cleveland	X	Ensemble	93%	X
15	Guarneros et al. [50]	Cleveland	X	LR	77.22%	X
16	Rahim et al. [44]	Cleveland	SM	MaLCaDD	95.5%	X
17	**Proposed Work (HXAI-ML)**	**Cleveland**	**RO+TL**	**ETC**	**98.78%**	**PIA, SHAP, LIME**

Gupta et al.'s RF (97.13%) and Suhatril et al.'s RF (85%), which performed significantly lower. The exceptional results on the Framingham dataset further reinforce the potential of HXAI-ML as a state-of-the-art tool for heart disease prediction, offering a combination of high accuracy, advanced data balancing techniques, and explainability—making it a promising solution for clinical decision support.

The improved performance of HXAI-ML can be attributed to its unique combination of data balancing techniques (RO+TL), along with the use of the ETC, which enhanced both prediction stability and accuracy. Additionally, the inclusion of Explainable AI methods such as SHAP, LIME, and PIA helped in making the model's predictions more understandable and trustworthy, which is crucial in clinical decision-making contexts. The results on the multiple datasets further underscore the effectiveness of HXAI-ML in heart disease prediction, positioning it as an advanced tool in the healthcare domain that combines high accuracy with explainability.

Table 12
Comparison Analysis of our Proposed (HXAI-ML) Model with existing works on Framingham Dataset.

SI. No.	Author	Dataset	Data Balancing	Model	Accuracy	XAI
1	Mienye et al. [27]	Framingham	X	XGBoost	95.9	SHAP
2	Kayalvizhi et al. [41]	Framingham	X	CNN-BiLSTM	95.3%	X
3	Mienye et al. [35]	Framingham	X	Ensemble	91%	X
4	Kahouadji [42]	Framingham	ROSE	BDT	85%	X
5	Omotehinwa et al. [47]	Framingham	X	LGBM	98.82%	X
6	Zhang et al. [49]	Framingham	X	LGBM	83.40%	X
7	Suhatril et al. [48]	Framingham	X	RF	85%	X
8	Sunge et al. et al. [45]	Framingham	X	LR, GB	85%	SHAP
9	Demir et al. [54]	Framingham	X	ANN	85.85%	X
10	Gupta et al. [53]	Framingham	X	RF	97.13%	X
11	Guarneros et al. [50]	Framingham	X	LGBM	77.42%	X
12	Louridi et al. [43]	Framingham	SM	Stacking	90.24%	X
13	Rahim et al. [44]	Framingham	SM	MaLCaDD	99.1%	X
14	Nalluri et al. [51]	Framingham	X	LR	85.86%	X
15	Gupta et al. [46]	Framingham	SM	SVM	83%	X
16	Beunza et al. [52]	Framingham	ROSE	LR	69%	X
17	**Proposed Work (HXAI-ML)**	**Framingham**	**RO+TL**	**ETC**	**99.24**	**PIA, SHAP, LIME**

5.1. Validation of research questions

- RQ1: The study evaluated combinations of RO, SM, RO+TL, RO+IHT, SM+TL and SM+IHT with ML models. Among these, the RO+TL combinations with ML demonstrated superior performance, effectively mitigating class imbalance and enhancing model generalization. The RO+TL+ETC technique yielded the highest accuracy of 98.78% for Cleveland dataset and 99.24% accuracy for Framingham dataset, demonstrating its ability to balance sensitivity and precision effectively (Table 3).
- RQ2: The integration of XAI methods enabled detailed insights into feature importance and individual predictions. SHAP values highlighted the influence of key risk factors such as age, cholesterol levels, and blood pressure on model outcomes. LIME provided localized explanations for specific cases, facilitating clinical decision-making, while PIA ensured transparency by presenting patient-specific prediction rationales, bolstering the model's applicability in real-world scenarios (Fig. 9 and Fig. 11).
- RQ3: The hybrid XAI-based model achieved a significant improvement in predictive performance compared to traditional approaches. On the Cleveland dataset, it achieved an accuracy of 98.78%, sensitivity of 98.82%, and precision of 98.78%; On the Framingham dataset, it achieved an accuracy of 99.24%, sensitivity of 99.24%, and precision of 99.24% These results underscore the model's robustness and generalizability in accurately identifying both positive and negative CVD cases while maintaining interpretability (Table 5 and Table 6).

The validation of the research questions confirms that the proposed hybrid model, leveraging advanced data balancing techniques and XAI, effectively addresses the challenges of imbalanced datasets, enhances interpretability, and outperforms existing methods in terms of prediction accuracy and clinical relevance. Results reinforce its potential for integration into healthcare systems to improve cardiovascular disease prediction and management.

5.2. Implications for clinical practice

The results of this study have notable implications for clinical practice, particularly in enhancing the early detection and management of CVD. The hybrid RO+TL+ETC model demonstrated superior performance, including high accuracy, kappa, MCC, and f1-scores, which confirms its reliability in identifying at-risk individuals even in imbalanced datasets. This accuracy enables healthcare providers to implement timely interventions such as lifestyle changes, medical treatments, or increased monitoring, ultimately improving patient outcomes. By integrating robust data balancing techniques like RO and advanced TL, the model ensures adequate representation of minority classes in medical datasets, addressing a critical challenge in clinical applications where misclassification can have severe consequences. The model's high accuracy and low error rates suggest its potential as a reliable decision-support tool to complement traditional risk assessment methods, enhancing diagnostic precision. Additionally, the scalability and adaptability of this hybrid approach make it applicable to diverse healthcare datasets, supporting its broader use in various clinical settings. This methodology offers a pathway to more intelligent, data-driven decision-making processes, fostering early intervention and better management of CVD, while also serving as a framework for addressing other complex health conditions.

5.3. Novelty of the proposed HXAI-ML framework

The novelty of the proposed Hybrid Explainable Artificial Intelligence Machine Learning (HXAI-ML) framework lies in its integration of advanced data balancing techniques, explainable AI methods, and high-performance ML models to address critical challenges in CVD prediction. The following innovative aspects distinguish this study from prior work:

Hybrid data balancing approach The framework introduces a novel combination of Random Oversampling (RO) with Tomek Links (RO+TL), addressing class imbalance more effectively than traditional methods like SM.RO ensures the dataset is balanced, while TL removes overlapping instances near decision boundaries, reducing noise and improving model generalization. This hybrid method has not been extensively explored in CVD prediction and demonstrates superior performance across two benchmark datasets.

Integration of explainable AI techniques Unlike prior works that primarily focus on black-box ML models, the HXAI-ML framework incorporates three complementary XAI methods: SHAP (Shapley Additive Explanations), LIME (Local Interpretable Model-agnostic Explanations), and Permutation Importance Analysis (PIA). This combination allows for both global interpretability (understanding overall feature contributions) and local interpretability (explaining individual predictions), enhancing clinical trust and enabling actionable insights for healthcare professionals.

Advancements in model selection and optimization The framework leverages the Extra Trees Classifier (ETC), an ensemble learning method known for its robustness and efficiency. Paired with the RO+TL balancing technique, ETC achieves state-of-the-art performance on the Cleveland and Framingham datasets, surpassing other commonly used classifiers like Random Forest and XGBoost. This optimized combination

ensures high accuracy (e.g., 98.78% on Cleveland) while maintaining computational efficiency.

Addressing CVD-specific challenges The HXAI-ML framework is explicitly tailored for CVD prediction, addressing three critical challenges:

- **Class Imbalance:** Through the novel RO+TL technique, the framework improves model performance on minority classes, ensuring fair predictions for high-risk CVD cases.
- **Interpretability:** By integrating XAI tools, the framework overcomes the "black-box" limitation of ML models, allowing healthcare practitioners to understand and trust predictions.
- **Scalability and Adaptability:** The framework is designed to be dataset-agnostic, achieving consistent performance across multiple datasets, and highlighting its scalability to diverse clinical settings.

Comprehensive experimental validation The framework is validated on two benchmark datasets (Cleveland and Framingham) and evaluated using multiple performance metrics. The experiments demonstrate that the combination of RO+TL with ETC achieves superior accuracy, precision, and interpretability compared to existing approaches. For instance, the framework achieves 99.24% accuracy on the Framingham dataset, showcasing its robustness.

Clinical applicability Unlike prior models that primarily aim for accuracy, the HXAI-ML framework ensures predictions are interpretable and actionable. By providing detailed insights into the most influential features (e.g., age, cholesterol, smoking), the framework empowers clinicians to make informed decisions and prioritize early interventions.

Contribution to the field The HXAI-ML framework contributes to the field by:

- Bridging the gap between performance and interpretability in ML-based CVD prediction.
- Introducing a novel, hybrid balancing technique (RO+TL) that enhances generalization and addresses imbalanced datasets effectively.
- Providing a clinically relevant, interpretable solution that aligns with the growing emphasis on transparency and trust in AI applications in healthcare.

By explicitly outlining these innovative aspects, the HXAI-ML framework sets itself apart from prior studies and demonstrates its significance in advancing the application of ML and explainable AI in cardiovascular disease prediction.

Limitation of the proposed work Despite the promising results, this study has some limitations. First, we did not explore deep learning (DL) techniques, which could improve model performance by capturing more complex patterns in the data. Second, the study did not investigate feature fusion methods, which might have enhanced the model's ability to integrate diverse data sources. Finally, transformer-based approaches, known for their effectiveness in handling sequential data, were not considered in this work.

Future works While our approach shows great promise, there are still areas for improvement. Future work will explore the integration of DL methods to capture more complex patterns, investigate feature fusion techniques for better data integration, and consider transformer-based models to handle sequential data more effectively. This will further enhance the model's performance, contributing to more accurate and clinically applicable heart disease detection systems.

6. Conclusions

This study presents a novel hybrid ML model for accurate prediction of CVDs, addressing challenges such as data imbalance and model interpretability. The proposed model integrates advanced data balancing techniques with XAI methods, offering both high accuracy and clinical interpretability. We combine oversampling and undersampling techniques, such as RO, SM, RO+TL, SM+TL, RO+IHT and SM+IHT, to balance the dataset and applied ML models to improve models performance. Additionally, XAI methods, including PDP, SHAP, and LIME, are incorporated to provide transparent and interpretable predictions, which are critical for clinical applications. The hybrid model's effectiveness is evaluated on two benchmark CVD datasets: Cleveland and Framingham, utilizing several ML algorithms such as RF, DT, GB, XGB, and ETC. Our results show that the combination of RO+TL with ETC achieves the highest performance, with an accuracy of 98.78%, precision of 98.82%, and F1-score of 98.77% using Cleveland dataset and 99.24% accuracy, precision and f1-score. The findings highlight the proposed model's potential to improve CVD prediction accuracy while ensuring interpretability for healthcare professionals.

7. Abbreviation

Table 13 summarizes key abbreviations used in the paper, covering terms related to heart disease detection, ML models, data balancing techniques, and explainable AI methods. It serves as a reference to assist readers in understanding the abbreviations in the methodology and analysis sections.

Table 13
Abbreviations and their Full Forms.

Abbreviation	Full Form
CVD	Cardiovascular Disease
CHD	Coronary Heart Disease
WHO	World Health Organization
FHS	Framingham Heart Study
ML	Machine Learning
XAI	Explainable Artificial Intelligence
RO	Random Oversampling
SM	Synthetic Minority Oversampling
TL	Tomek Links
IHT	Instance Hardness Threshold
RFC	Random Forest Classifier
DTC	Decision Tree Classifier
GBC	Gradient Boosting Classifier
XGBC	XGBoost Classifier
SVM	Support Vector Machine
LR	Logistic Regression
BDT	Boosted Decision Tree
XGB	eXtreme Gradient Boosting
RF	Random Forest
CART	Classification and Regression Tree
WACE	Weighted Aging Classifier Ensemble
ANN	Artificial Neural Network
LightGBM	Light GB Machine
OCT	Optimal Classification Trees
SHAP	Shapley Additive Explanations
PIA	Permutation Importance Analysis
LIME	Local Interpretable Model-agnostic Explanations
AdaBoost	Adaptive Boosting
PCA	Principal Component Analysis
KNN	k-Nearest Neighbors
GWO	Grey Wolf Optimization
AB-WAE	Accuracy-Based Weighted Aging Ensemble
NB	Naive Bayes
NN	Neural Networks
GB	Gradient Boosting
MLP	Multi-Layer Perceptron
CatBoost	Categorical Boosting
QPSO	Quantum-behaved Particle Swarm Optimization

(*continued on next page*)

Table 13 (*continued*)

Abbreviation	Full Form
MaLCaDD	Machine Learning Classifier with Data Balancing
QNN	Quantum Neural Network
MLP	Multi-Layer Perceptron
CNN	Convolutional Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
NRO	Newton-Raphson Optimizer
SFS	Sequential Feature Selection
RNN	Recurrent Neural Network
MICE	Multivariate Imputation Chained Equations
EHMFFL	Ensemble Heuristic–Metaheuristic Feature Fusion Learning
GA	Genetic Algorithm
FRS	Framingham Risk Score
KMC	K-Means Clustering
PCA	Principal Component Analysis
PIAr	Pairs of instances at risk
AUC	Area Under Curve

CRediT authorship contribution statement

Md. Alamin Talukder: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amira Samy Talaat:** Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Mohsin Kazi:** Visualization, Validation, Software, Resources, Project administration, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Availability of data and materials

The selected datasets are sourced from free and open-access sources, such as: Cleveland CVD Dataset: https://archive.ics.uci.edu/static/public/45/heart+disease.zip and Framingham CVD Dataset: https://www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset.

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